



Stroke Risk Prediction and Prevention: Traditional versus Machine Learning Approaches

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Received: 10 July 2025 / Revised: 6 September 2025 / Accepted: 12 September 2025

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Abstract

Stroke is a leading cause of disability and mortality globally, imposing a significant impact on individuals, healthcare systems, and society. Existing diagnostic techniques for stroke prediction, including risk scores and imaging assessments, frequently lack the accuracy required for personalised treatment. Conversely, machine learning (ML) and deep learning (DL) models provide the capability for enhanced accuracy and personalised predictions by employing complex high-dimensional data sets. However, these methodologies experience challenges such as generalisability, interpretability (blackbox ML algorithm), and data quality. This systematic review, which does not strictly adhere to all PRISMA guidelines, aims to provide an extensive overview of stroke prediction methods by offering transparent methodology descriptions and employing a PICO-based search strategy to analyse studies from 1981 to 2024, focussing on stroke prediction, diagnosis, prognosis, and treatment. A total of 3064 articles were initially identified from major databases (PubMed, Medline, EMBASE), with 226 studies included following a thorough screening process. The review integrated findings from traditional clinical instruments and modern AI methodologies, highlighting data types (imaging, signal, and tabular), biomarkers, model efficacy, and clinical relevance. Machine learning and deep learning models displayed enhanced predictive accuracy and capability for personalised stroke risk classification, especially when incorporating multimodal data, including neuroimaging, electrocardiograms, and biomarkers such as plasma fibrinogen. Advanced models such as CNNs and LSTMs demonstrate superior performance compared to traditional tools in the processing of complex data; however, they are constrained by issues related to generalisability and interpretability. Classical tools continue to hold significance, particularly in low-resource environments and initial stroke evaluations. Hybrid approaches that integrate machine learning with traditional methods demonstrate potential for improved clinical utility. AI-driven methods indicate a notable development in stroke prediction; however, their incorporation into clinical practice requires the resolution of critical limitations, including model transparency, data diversity, and ethical considerations. Future research must emphasise explainable AI, multimodal data fusion, and equitable access to guarantee that machine learning-enhanced tools are accurate, interpretable, and scalable across varied patient populations. Such initiatives are essential for enhancing stroke prevention, diagnosis, and management globally.

1 Introduction

Stroke is a significant worldwide public health concern, being one of the primary contributors to mortality and permanent disability on a global scale. The World Health Organization (WHO) estimates that around 15 million individuals experience stroke each year, with around 5 million leading to mortality and an additional 5 million resulting in permanent disability [214, 215]. The high frequency and significant impacts of stroke emphasize the need for adequate prevention, precise prediction, and effective management approaches. The increasing age of the world population is projected to lead to a higher prevalence of stroke,

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thereby requiring improvements in both clinical and technology methods for managing strokes.

Traditionally, the diagnosis and treatment of strokes placed a high value on well established clinical evaluations, imaging methods, and standardized scales, such as the National Institutes of Health Stroke Scale (NIHSS) [68, 211] and the Glasgow Coma Scale (GCS) [190]. These methodologies have played a vital part in directing medical opinions and treatment outcomes. However, the limitations of these methods have become more evident, notably in their ability to accurately predict stroke occurrences and tailor treatment for various groups of patients. Conventional approaches often come up short in comprehensively capturing the complex nature of specific patient profiles, ultimately resulting in less than ideal results.

Despite such limitations, there has been a notable increase in interest within the medical community regarding the use of ML and artificial intelligence (AI) in managing strokes. ML models are specifically optimized to examine extensive and intricate datasets, enabling the detection of patterns and correlations that may not be visible using traditional statistical approaches. This capability has established ML as a potentially revolutionary tool in the prediction and treatment of strokes. Thus, ML algorithms have the capability to combine and examine data from many origins, including patient history, imaging, and biomarkers, in order to achieve more precise and personalized predictions of stroke risk and recovery results.

A block diagram depicting data processing pipeline, demonstrating how different types of inputs are transformed into outcomes through a variety of modeling techniques is presented in Fig. 1. Data in multiple formats, such as pictures, signals, and tabular data, make up the inputs. These inputs serve as the raw materials for analysis. These inputs are processed using different modeling techniques, which include statistical models, ML algorithms, DL architectures, and hybrid approaches. Every kind of modeling, from straightforward statistical analysis to intricate neural networks that can process unstructured data, has specific advantages. The outputs represent three types of outcomes. Descriptive outcomes focus on understanding what has happened, predictive outcomes forecast future events based on data, and prescriptive outcomes recommend optimal courses of

action based on predicted results. This diagram illustrates the process from unprocessed data to insights, highlighting the crucial role that models play in converting a variety of data formats into meaningful knowledge. It emphasizes how many modeling techniques can be used based on the kind of data and the intended result, offering a flexible framework for making decisions based on data.

Whereas ML shows considerable potential in enhancing the management of strokes, there are still some challenges and limitations that exist. The generalizability of ML models across disparate populations is a major challenge. A large percentage of ML models are trained on datasets that do not contain significant diversity, resulting in biased predictions that may not be relevant to all patient populations [217]. In addition, the interpretability of ML models poses a substantial challenge. Although these models have the capability to generate very precise predictions, the opaque nature of many algorithms poses challenges for physicians in comprehending and relying on the decision making process, therefore may restrict their incorporation into clinical practice [32]. Furthermore, there are challenges regarding the cost efficiency and sustainable application of these technological advances, especially in settings with limited resources where access to the required infrastructure may be limited.

Addressing these challenges, the objective of this work is to offer a thorough evaluation of the current condition of stroke prediction and treatment, specifically by comparing conventional methods with ML-based techniques. The objective of the review is to discover the advantages and drawbacks of each method, based on primary discoveries from recent research. Through this approach, it tries to highlight those areas in which ML models have demonstrated substantial potential, as well as identifying their limitations.

Emphasizing both the advances that ML has brought and the important limitations that follow its integration into clinical practice, this paper offers a thorough assessment of stroke prediction and management techniques. Among the most remarkable contributions is the proof that ML models (especially DL approaches like CNNs and LSTMs) exceed conventional tools in processing complicated, high-dimensional data to forecast stroke results with more accuracy [218, 221]. By including patient-specific information and pertinent biomarkers like plasma fibrinogen and urinary metabolites, which improve diagnostic accuracy and prognosis accuracy, these models clear the path for more individualized, data-driven treatment. The study, on the other hand, points out some restrictions that qualify these encouraging results. The generalizability of ML models (which tend to perform well in controlled settings but exhibit lower accuracy in varied, real-world populations owing to

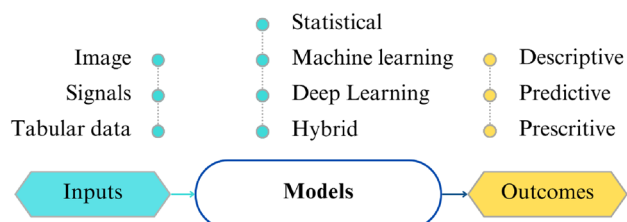


Fig. 1 A block diagram depicting transformation of input data into actionable outcomes

demographic, comorbidity, and stroke subtype variation) is a key worry.

Additionally, in resource-constrained environments, the dependence on huge, high-quality data sets becomes a hurdle; for healthcare professionals without specialized training, the intricacy of ML models creates difficulties with interpretation and acceptance. The results also warn against undervaluing conventional techniques, which are still useful particularly in mild to moderate stroke situations and are more readily used across different clinical settings. Therefore, the work supports a hybrid strategy that combines ML with conventional techniques, therefore combining the accuracy of data-driven models with the accessibility and clinical validation of older approaches. It underlines even more the need of future studies meant to increase data diversity, create explainable artificial intelligence, revise clinical training, and guarantee ethical, fair, and reasonably priced use of ML in healthcare.

This study specifically analyzes the relationship between traditional diagnostic tools and machine learning methodologies, unlike other evaluations that largely focused on one or the other in isolation. Through a comprehensive comparison of traditional risk scores, imaging modalities, and clinical scales with modern machine learning and deep learning models, we highlight their respective advantages and drawbacks, as well as the specific settings in which each methodology is most effective. This comparative analysis highlights the enduring significance of conventional techniques in low-resource or preliminary diagnostic environments, while also illustrating the novel capabilities of machine learning in managing intricate, multimodal data. This review highlights the complementarities of various paradigms and presents a framework for hybrid approaches that merge data-driven innovations with conventional therapeutic procedures, thus providing a unique contribution to the existing research.

Furthermore, this study will examine the present gaps in the research, including the requirement for a wider range of various and comprehensive datasets, the development of comprehensible ML models, and the evaluation of cost effectiveness in various healthcare environments. The evaluation suggests future research topics and approaches to address these gaps, which may help overcome the existing restrictions and improve the academic discipline. Some of these suggestions are to use fairness aware algorithms [155] to reduce bias, create multimodal DL models that combine different types of data, and run longitudinal studies to improve the accuracy of predicting long term results.

To guide this review, several key research questions are posed:

RQ1. What is stroke, and what are its global health impacts, associated disorders, and primary risk factors?

RQ2. What data sources, categories of data, and study designs are commonly employed in stroke research?

RQ3. What tools, technologies, and analytical methodologies are used in stroke research, and what are their strengths and limitations?

RQ4. What gaps exist in current stroke research, and what directions should future studies pursue?

The remainder parts of this work can be categorized based on their individual sections as follows: Sect. 2 focuses on the history and facts about stroke and types of studies on stroke are mentioned in Sect. 3. Section 4 discusses the methodologies we used in our study. Section 5 provides a comprehensive analysis of dataset, Sect. 6 discusses the key risk factors associated with stroke, Sect. 7 highlights tools and techniques employed to study stroke and performance evaluation used. In Sect. 8, we discuss key findings, comparative analysis of different approaches, identifying gaps and future research directions followed by conclusion in Sect. 9. Definitions of specialized medical and artificial intelligence terminology referenced in this paper can be found in Appendix A, Table 8.

2 Background

Stroke has been acknowledged as a fatal medical disorder since ancient times, with first accounts tracing back to ancient Greek doctors who called it “apoplexy” [153]. Throughout the years, knowledge of stroke has developed, especially with progress in medical science. In the 20th century, stroke was categorized into ischemic and hemorrhagic disorders, depending on whether the stroke resulted from an occluded artery or a ruptured blood vessel. Research has subsequently broadened its scope to primarily investigate stroke risk factors, prevention, and treatment. Investigations on stroke encompass a wide spectrum, ranging from observational cohort studies that monitor the occurrence and consequences of strokes to randomized controlled trials that assess therapies such as thrombolysis and surgical procedures. Contemporary research also encompasses the identification of biomarkers, genetic investigations, and the use of ML methods to forecast results and enhance treatment approaches, so enhancing our comprehension of this intricate disease.

2.1 History and Facts

A stroke (cerebrovascular accident) is a severe medical disorder caused by an abrupt cessation or decrease in blood flow to a specific area of the brain, leading to a lack of oxygen and nutrients [117]. There are three primary classifications: ischaemic stroke, hemorrhagic stroke, and transient

ischaemic attack (TIA). Ischaemic strokes are the most common, resulting from blocked blood vessels, with Embolic Stroke of Undetermined Source (ESUS) as an important subtype [66]. Hemorrhagic strokes occur when a blood vessel ruptures, while TIAs involve temporary blockages that serve as warning signs for future strokes [89]. Stroke is the second leading cause of death and the third leading cause of disability worldwide, accounting for nearly six million deaths annually. In 2021, there were about 70 million cases of ischaemic stroke globally, with a sharp rise since 1990. In India, stroke is the fourth leading cause of mortality, with a higher burden in rural areas lacking adequate healthcare infrastructure [42]. Low- and middle-income countries (LMICs) represent the majority of global stroke-related deaths and disabilities. Major risk factors include hypertension, diabetes, high cholesterol, smoking, obesity, and sedentary lifestyle. Symptoms may involve sudden weakness, impaired speech, visual disturbances, dizziness, and severe headaches [117]. The condition also imposes lasting physical, cognitive, and emotional challenges for survivors and families.

2.2 Stroke and Other Disorders

Findings from a compilation of studies have revealed the complex network of factors that contribute to the occurrence of strokes. In addition to the conventional risk factors commonly linked to stroke, such as hypertension and diabetes, these investigations have revealed a wide range of varied contributors. Studies examining heart attacks, levels of fibrinogen in the blood, and disorders such as irregular heart rhythm have uncovered intricate links to the occurrence of strokes, emphasizing the complex nature of this type of brain blood vessel event. Furthermore, the investigation of psychological elements, such as stress, sadness, and social isolation, yielded convincing observations about their impact on stroke vulnerability and recovery results. In addition, acknowledging the importance of familial history and hereditary variables as substantial contributors to the risk of stroke enhances our comprehension of this intricate disorder.

2.2.1 Diabetes

Diabetes greatly heightens the likelihood of stroke through various ways. Elevated blood glucose levels lead to vascular damage, facilitating the progression of cholesterol and constriction of the arteries that provide blood to the brain [1]. People with diabetes have a higher susceptibility to conditions such as atrial fibrillation and coronary artery disease, which in turn increases the risk of stroke [31].

2.2.2 Psychosocial

Psychosocial issues, such as depression, have a substantial impact on both the likelihood of experiencing a stroke and outcomes after a stroke. People suffering from depression frequently display harmful lifestyle habits and physiological alterations that heighten the likelihood of stroke, including inflammation and impaired endothelial function [10, 40, 87].

2.2.3 Left Atrial Size

Left atrial enlargement is associated with a higher likelihood of stroke. Enlargement of the left atrium can increase the possibility of developing atrial fibrillation, a condition characterized by an abnormal heart rhythm that can result in the production of blood clots [19]. Evaluating the dimensions of the left atrium using imaging methods like echocardiography can aid in the identification of individuals with an increased likelihood of experiencing a stroke [121, 180].

2.2.4 Myocardial Infarctions

Myocardial infarctions, often known as heart attacks, pose a substantial risk for stroke [79]. Myocardial infarction occurs when there is a clog in the walls of the coronary arteries, causing a decrease in the flow of blood to the heart muscle, hence heightening the likelihood of a subsequent stroke [122, 161]. In addition, persons who have had a heart attack may have preexisting cardiovascular disorders which increase their chances of having a stroke [142].

2.2.5 Plasma Fibrinogen

Plasma fibrinogen is a protein that helps blood clot. High levels of this protein have been linked to a higher chance of stroke [79, 122]. Hence, monitoring of plasma fibrinogen levels can offer useful insights into evaluating the risk of stroke and applying preventative interventions.

2.2.6 Blood Pressure

Hypertension, often known as high blood pressure, is a major contributing factor to the occurrence of stroke [75, 126]. Prolonged hypertension can cause gradual deterioration of arterial walls, rendering them more vulnerable to the accumulation of lipid deposits (plaques) and thrombus development. These plaques have the potential to constrict the arteries, leading to a decrease in blood flow to the brain [126, 136, 141].

2.2.7 Atrial Fibrillation

Atrial fibrillation (AF) significantly increases the risk of ischemic stroke, but the relationship is complex and not causal. AF may indicate an underlying atrial cardiopathy that promotes thrombosis, however not all thromboembolic events occur temporally with AF episodes. This includes atrial remodeling, left atrial appendage blood stasis, and endothelial dysfunction. Atrial fibrosis, common in atrial fibrillation patients, has been linked to stroke risk even in those without arrhythmia. These findings suggest that atrial fibrillation is both a trigger and a clinical presentation of a large pathological substrate that predisposes people to stroke. AF is a substantial indirect risk factor, indicating atrial illness with embolic potential, and should be included in modern risk categorization models and anticoagulant treatment [27, 28, 48, 145].

Summary: The recent studies highlight some of the complex and varied nature of the risk of stroke. Medical disorders such as diabetes mellitus and myocardial infarctions, as well as biomarkers like plasma fibrinogen, each have a significant impact on an individual's vulnerability to stroke. In addition, psychosocial factors such as depression and physiological markers such as left atrial size offer further understanding of the risk and consequences associated with stroke.

3 Types of Studies

In recent decades, stroke has become a global health issue due to its high morbidity, death, and socioeconomic effect. A significant amount of study has been focused on comprehending the various elements of stroke, including its epidemiology, pathophysiology, risk factors, clinical presentation, acute care, rehabilitation strategies, and long term consequences. This review synthesizes key findings and advances in the field to explain current knowledge and identify future research trends and priorities. There are four critical areas of stroke related research: reviews, prediction, prognosis/outcome, and treatment. Together, these categories underscore the comprehensive approach necessary for effective stroke management and prevention.

3.1 Review

Stroke research has made substantial progress in several areas, such as prognostic indicators, diagnostic biomarkers, meta analyses, predictive parameters, perioperative stroke prevention, and ML based prediction models. Prognostic studies have identified age, stroke severity, and comorbidities as crucial determinants of stroke outcomes, providing

guidance to doctors in customizing treatment and rehabilitation programs [43]. Plasma fibrinogen and other blood based markers are diagnostic biomarkers that assist in the early diagnosis of strokes and the classification of risk [208]. Meta analyses combine data from several research to offer complete insights into the risk factors, effectiveness of treatment, and outcomes of strokes [209]. Neuroimaging data and genetic markers are predictive indicators that can be used to identify individuals who are at a greater risk of experiencing stroke recurrence or consequences. Implementing strategies for perioperative stroke prevention, such as employing refined surgical methods and administering appropriate pre-operative medical care, can effectively reduce the likelihood of stroke occurring during surgical procedures [134]. ML methods utilize extensive datasets to improve the prediction of stroke risk, providing individualized risk assessment and informing tactics for prevention [32].

Several types of review studies have examined stroke prediction, diagnosis, and prognosis, emphasising the transition from traditional methods to ML based methodologies. Table 1 presents a comparative overview of five significant researches. Initial evaluations of prognostic models and diagnostic biomarkers revealed insufficient model validation and a lack of dependable biomarkers, compounded by issues such as variance in data and inconsistent validation processes. Cohort-based studies, such as those utilising the Framingham data, validated the significance of conventional risk factors but were constrained by outdated model frameworks. Narrative evaluations regarding perioperative stroke risk provided valuable clinical insights but were deficient in quantitative synthesis. Recent evaluations of ML technologies exhibited enhanced prediction accuracy and personalisation compared to traditional methods. Nevertheless, generalisability and external validation continue to be significant limits. These results underscore the necessity for powerful, adaptable, and clinically relevant stroke prediction algorithms.

3.2 Prediction

Recent research have underlined important developments and results in the prevention and identification of stroke. While atorvastatin considerably lowers stroke risk, supporting its use in diabetic patients and stressing the need of considering diabetes specific factors in stroke risk assessments, the Collaborative atorvastatin diabetes study (CARDS) trial showed that elements like age, microalbuminuria, and poor glycemic control increase stroke risk in Type 2 diabetes patients [82]. Future studies might use other data sources to improve accuracy. Leading cause of death and disability, ischemic stroke (IS), has become much more burdensome; primary contributing reasons include smoking and excessive

Table 1 Comparison of review studies related to stroke prediction and risk assessment

Study (year)	Main focus	Methodology	Key findings	Strengths	Limitations
[43] (2001)	Prognostic models post-stroke	Systematic review of 83 models	Poor validation of models	Comprehensive quality assessment	No meta-analysis, data heterogeneity
[208] (2008)	Diagnostic biomarkers for stroke	Systematic review (58 biomarkers)	No clinically reliable biomarker	Evaluates accuracy (sensitivity/specificity)	No consistent validation or blinding
[209] (2008)	Coronary heart disease & stroke event prediction	Accelerated failure time models on Framingham cohort	Traditional risks + Body mass index effective	Large dataset, long follow-up	Traditional model structure
[134] (2018)	Perioperative stroke risk	Narrative literature review	Lists clinical and surgical predictors	Broad practical insights	No quantitative synthesis
[32] (2023)	ML-based stroke prediction & stratification	Review of ML tools (e.g., Deep neural network, Random forest) for stroke prediction	ML > traditional models for Cardiovascular disease (CVD)/stroke	High predictive accuracy, personalization	Need for external validation, generalizability issues

BMI. Reducing the worldwide load of IS depends on good regulation of several elements [60]. Another study indicates that permeability magnetic resonance imaging (MRI) can be used to evaluate early blood brain barrier abnormalities in acute ischemic stroke, so perhaps enhancing treatment options and outcomes by spotting patients more likely of hemorrhagic change [203]. Detecting embolic signals with transcranial Doppler (TCD) can predict stroke risk in individuals with asymptomatic carotid stenosis, according the Asymptomatic carotid emboli study (ACES), therefore improving the selection procedure for carotid endarterectomy [123]. Furthermore validated for early stroke recurrence, the Recurrence risk estimator (RRE) score shows great success for focused stroke prevention in clinical environments [5]. Finally, include biomarkers of carotid plaque vulnerability—such as volume, neovascularization, and inflammation—could transform stroke preventive programs and go beyond depending just on stenosis degree [167].

Recent advances in computational physiology are revolutionizing stroke risk prediction in atrial fibrillation (AF) by moving beyond traditional scoring systems such as CHA₂DS₂-VASc. Using tools such as computational fluid dynamics (CFD), researchers are identifying key hemodynamic risk factors (such as low wall shear stress and blood stasis) within the left atrium and appendage, areas prone to clot formation. Studies have shown that incorporating atrial motion [124], fibrosis mapping [147], and functional imaging [226] improves understanding of individual stroke risk. The most advanced models [69] integrate electrophysiology, tissue mechanics, and blood flow, revealing how fibrosis impairs atrial function and promotes thrombogenesis.

3.3 Prognosis

Numerous advanced methodologies have been investigated in recent research to enhance the accuracy of outcomes predictions in patients with ischemic stroke. Perfusion and diffusion weighted MRI, which are advanced imaging techniques, have demonstrated potential in detecting cerebral ischemia and accurately predicting patient outcomes. These approaches are more effective than standard MRI methods, especially within the first six hours after the symptoms appear [205]. Research has shown that high levels of glucose in the blood can indicate a negative prognosis following a stroke. This indicates that hyperglycemia should be aggressively treated rather than being considered a simple stress response [206]. Meanwhile, echoplanar MRI techniques have proven to be useful in evaluating the progression of strokes by showing a strong correlation with clinical outcomes and the final size of the damaged area in the brain. This can potentially help guide treatment approaches [17]. The Alberta Stroke Programme Early CT Score (ASPECTS) has demonstrated its reliability as a tool for predicting functional outcomes and the probability of symptomatic intracerebral hemorrhage. It has shown superior performance when compared to certain prognostic scores [18]. Moreover, the JURaSSiC study emphasized the enhanced precision of the iScore in comparison to physicians' forecasts regarding stroke outcomes, indicating that it could be a more dependable instrument for making decisions about acute stroke therapy [171]. ML algorithms, namely deep neural networks (DNN), have demonstrated enhanced predictive accuracy for long term outcomes in stroke patients when compared to traditional scoring systems. This suggests that these technologies have the potential to improve clinical decision making [80]. In order to enhance the accuracy of predictive models and gain a deeper understanding of these findings, it is important to execute further research utilizing larger study groups and more advanced imaging techniques.

Table 2 Comparative overview of predictive approaches and treatment strategies in stroke management research

Category	[194]	[4]	[138]
Objective	Identify AF patients at low stroke risk when on aspirin (to guide anticoagulation)	Predict acute and total hospital length of stay (LOS) for stroke patients	Predict final infarct volume and assess treatment effect using DL
Population	2501 patients with nonvalvular atrial fibrillation on aspirin (from 6 trials)	295 first-ever stroke patients who survived initial hospitalization (Sweden)	222 ischemic stroke patients (187 treated with rtPA) from multi-center studies
Prediction Method	Recursive partitioning algorithm (clinical prediction rule)	Multivariate regression analysis	Deep convolutional neural networks (CNNdeep, CNNshallow, etc.)
Main Predictors	Previous stroke/TIA, hypertension, CAD, diabetes	NIHSS score, prestroke ADL, dementia, stroke subtype (lacunar, posterior)	9 acute MRI biomarkers: ADC, DWI, PWI-derived maps (Tmax, CBV, CBF, etc.), T2-FLAIR
Key Outputs	Low vs. high stroke risk category	Acute LOS (mean 12 days), Total LOS (mean 29 days)	Infarct risk maps, infarct volume prediction, and differentiation of treatment effect
Treatment Assessed	Aspirin vs. oral anticoagulation (warfarin)	Standard stroke care + varied rehab pathways	rtPA (IV thrombolysis); model assessed impact of treatment by comparing predicted vs. actual infarct volume
Treatment Findings	Low-risk patients on aspirin had event rates similar to general population; little added benefit from anticoagulation	LOS impacted by prestroke functional state and severity, but treatment type not directly analyzed	CNNdeep showed treatment effect: smaller predicted infarcts in rtPA-treated patients vs. non-treated ($P=0.048$)
Clinical Usefulness	Avoid unnecessary anticoagulation in low-risk AF patients	Support planning for rehabilitation and discharge needs	Personalized treatment planning; potential for automated decision support in acute stroke

3.4 Treatment

Timely and efficient treatment following a stroke is essential for enhancing patient results and maximizing health-care resources. A study was conducted with the objective of creating a clinical prediction rule to identify persons with nonvalvular atrial fibrillation (AF) who are taking aspirin and have a low risk of stroke. This could potentially eliminate the requirement for oral anticoagulation. The study analyzed data from patients with AF and categorized those who did not have a history of stroke, untreated hypertension, symptomatic coronary artery disease, or diabetes as low risk. The findings revealed that anticoagulant medication had minimal advantages for these individuals [194]. A separate investigation aimed to develop a prognostic model that could precisely predict the duration of hospital stay for stroke patients. This was achieved by examining a group of individuals and taking into account several factors like the severity and kind of stroke, pre-existing dementia, smoking habits, and the level of dependent on daily tasks. The model exhibited exceptional precision, facilitating effective resource allocation and patient care [4]. In addition, a third study utilized acute magnetic resonance imaging to create a prognostic model for the ultimate size of lesions in patients with acute ischemic stroke. The study found that a deep convolutional neural network (CNNdeep) had better performance compared to previous techniques. This approach offers the potential to enhance automated decision support in stroke treatment, hence boosting the precision and customization of care regimens for individual patients [138]. These studies emphasize the significance of individualized methods in evaluating the risk of stroke, managing hospitalization, and planning treatment in order to enhance patient outcomes and efficiently use healthcare resources (Table 2).

Numerous studies investigated the function of anticoagulation and atrial cardiopathy in ESUS, which accounts for 20–30% of ischaemic strokes. Several randomised controlled trials have examined whether direct oral anticoagulants (DOACs) prevent subsequent strokes better than aspirin in ESUS patients. In the NAVIGATE ESUS study, rivaroxaban had a greater risk of serious bleeding than aspirin but no meaningful reduction in stroke recurrence [77]. The RE-SPECT ESUS trial comparing dabigatran to aspirin found no difference in stroke prevention, despite similar safety [53]. Due to the heterogeneity of ESUS, the ARCADIA study hypothesises that apixaban may benefit a subset of patients with atrial cardiopathy over aspirin [92]. Increasing observational and structural knowledge suggests that left atrial fibrosis, even without AF, may be a thrombotic substrate. Tandon et al. found that ESUS patients had atrial fibrosis on cardiac MRI similar to AF and significantly greater than healthy controls [61], while Bifulco et al. used

computational modelling to show that this fibrosis could sustain arrhythmogenic reentry patterns like AF [25]. Kühnlein et al. found that atrial fibrosis $\geq 12\%$ in ESUS patients was linked to increased risk of stroke recurrence or incident AF, suggesting that atrial cardiopathy should be included in future ESUS therapy strategies [102]. These findings suggest that stroke causes in ESUS may be more complex and that targeted anticoagulation based on atrial substrate features may provide personalised prevention.

Summary: To summarize, recent progress in stroke research has resulted in substantial enhancements in the prediction, identification, and management of the disease. Key variables like age, severity of stroke, and presence of other medical conditions, as well as diagnostic biomarkers such as plasma fibrinogen, are essential in the early detection and prediction of outcomes. Advanced meta analyses and neuro-imaging methods have yielded more profound understanding of stroke risk and recurrence, while ML models provide improved precision in forecasting stroke outcomes and directing individualized treatment. Preventive measures,

such as the administration of atorvastatin in diabetic patients and the use of noninvasive detection technologies, show potential in decreasing the risk of stroke. Moreover, the use of sophisticated imaging and prediction models enhances the process of making decisions and achieving improved treatment results, so emphasizing the capacity of developing technology to revolutionize the management of stroke.

4 Methodology

The UML activity flow diagram (Fig. 2) depicts the systematic and thorough methodology employed in this evaluation on stroke prediction and prevention. The study methods were carefully selected according to evidence-based research protocols, utilizing the PICO (Population, Intervention, Comparison, Outcome) architecture to ensure precision in defining research questions, search parameters, and inclusion criteria. These frameworks supervised the formulation of research topics, the selection of databases, and the implementation of inclusion and exclusion criteria, so ensuring transparency, replicability, and the mitigation of bias. The approach was upgraded by an extensive database search (PubMed, Medline, EMBASE, Scopus, Web of Science, CINAHL, CENTRAL), a dual-reviewer screening process, and systematic data extraction, all of which improved consistency and dependability. The review did not incorporate all components of the PRISMA 2020 framework, including protocol registration and the PRISMA flow diagram; however, it followed to established systematic review methodologies in a resource-efficient manner, aligning effectively with the study's objectives.

This study aims to objectively assess the significance of various methods and models in addressing various stroke related issues, such as prognostication, diagnosis, therapy, and risk factor identification in addition to prevention. Using a highly methodical methodology, this review thoroughly examines the most influential and methodologically sound studies in the subject. The Fig. 3 depicts the selection procedure for a systematic review of stroke research from 1981 to 2024. The method commences with a conceptual framework that delineates the problem and its four components: stroke prediction, stroke management, stroke treatment, and stroke prognosis, yielding 3064 articles. After first screening based on abstracts and titles, 2647 papers are eliminated for duplicity or irrelevance, leaving 417 for complete text review. A total of one 189 papers were retracted due to inappropriate study designs and/or outcomes, resulting in 226 articles included in the final analysis. The research seeks to offer a thorough overview of the contributions of techniques and models to stroke management by synthesizing the results from these important studies. Important findings

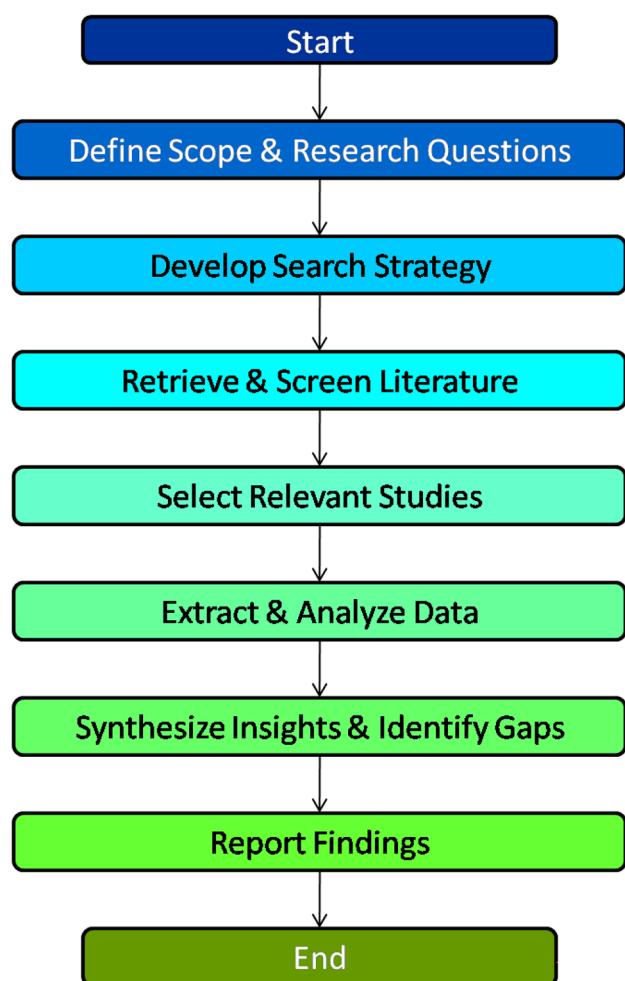
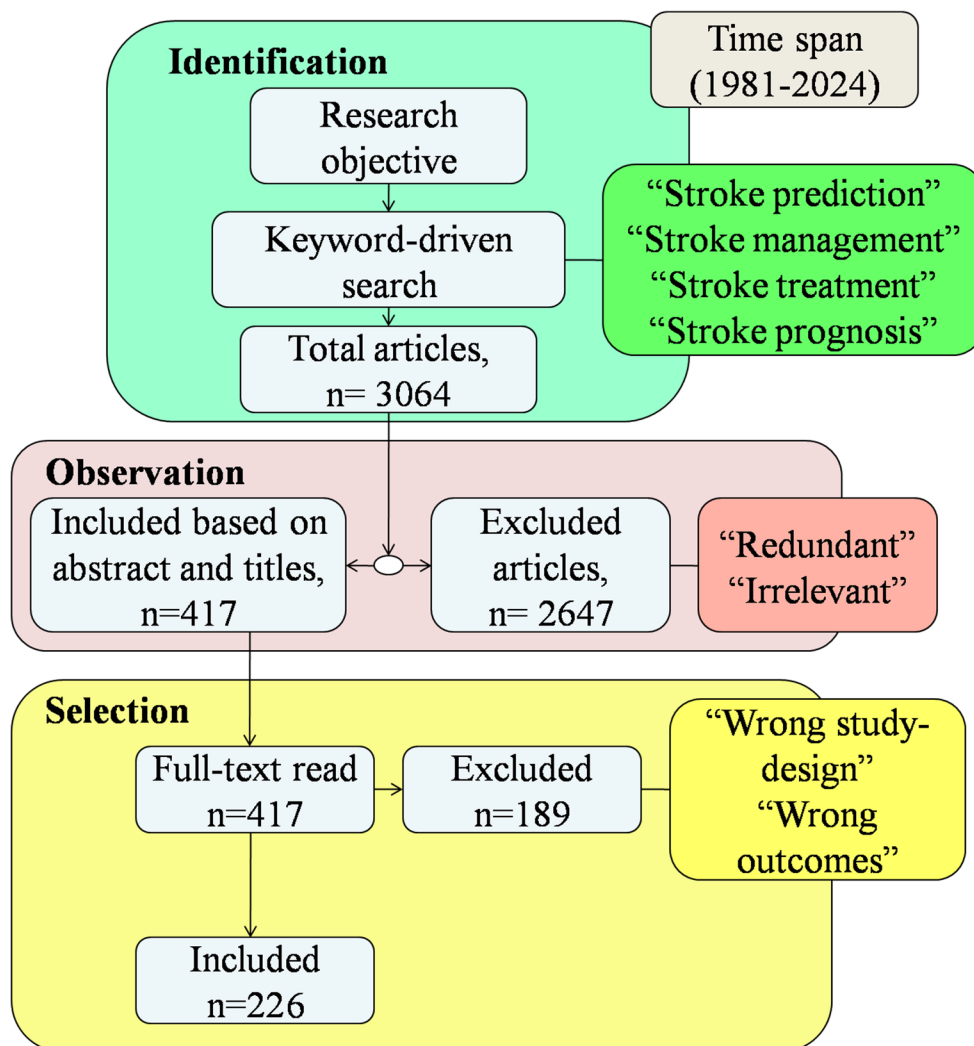


Fig. 2 Flowchart illustrating the operational steps of the research methodology

Fig. 3 Systematic review methodology for Stroke related studies (1981–2024)



from the literature will be highlighted in the discussion sections, providing insightful viewpoints that can influence and direct further study in this quickly developing field.

4.1 Search Strategy

A systematic approach was devised to conduct an extensive search on stroke prediction and management, covering material published from 1981 to 2024. A comprehensive search was conducted across prominent databases such as PubMed, Medline and EMBASE, Web of Science, Scopus, CINAHL, and the Cochrane central register of controlled trials (CENTRAL) to ensure a thorough coverage of pertinent studies. The search strings were constructed utilizing the PICO (Patient/Population, Intervention, Comparison, Outcomes) [173] framework to encompass the entirety of research pertaining to stroke prediction and management. The keywords encompassed a blend of broad terms such as “stroke”, “prediction”, and “management”, as well as more precise terms like “machine learning”, “regression

analysis”, “prognosis”, “rehabilitation outcome”, and “clinical efficiency”.

4.2 Study Selection

A thorough search strategy was executed to locate pertinent material on stroke prediction and management. Comprehensive searches were performed across the primary databases like PubMed, Medline, and EMBASE, utilizing specific keywords including “stroke prediction,” “stroke risk factors,” “machine learning,” and “stroke management.” Boolean operators (e.g., “AND,” “OR”) were utilized to enhance search accuracy and ensure the inclusion of studies pertaining to both predictive models and management approaches. To enhance searching, synonyms and alternative phrases were integrated, and the search syntax was tailored to the specific structure and requirements of each database. The database search, conducted on January 26, 2024, yielded 3064 articles: 985 from PubMed, 978 from Medline, and 1101 from EMBASE. Following the elimination of 237

duplicate entries, 2827 distinct records persisted. A preliminary screening eliminated 2410 publications considered extraneous to the study subject, resulting 417 articles selected for complete text review.

The inclusion criteria were limited to peer-reviewed studies published in English during the last thirty years that addressed empirical methods for stroke prediction or clinical management. Studies were rejected for lacking predictive analysis, not utilising original data, or failing to describe essential outcomes such as model performance or clinical efficacy. A manual quality assessment was performed for each study, evaluating the clarity of objectives, robustness of technique, appropriateness of sample size, and transparency of findings. Data extraction was conducted utilising structured Microsoft Excel sheets to document pertinent study parameters, encompassing author information, publication year, model type, predictor variables, validation procedures, and outcome measures, including sensitivity, specificity, and accuracy. Microsoft Word was utilized to arrange the retrieved data and provide narrative summaries. No specialised statistical software or programming techniques were utilized, and all analyses were purely descriptive. This systematic and resource-efficient approach guaranteed a comprehensive, transparent, and replicable examination of the current research on stroke prediction and management.

4.3 Screening

The screening method consisted of two stages: initially, a review of titles and abstracts, followed by a thorough examination of the full-text of the chosen papers. In first stage we eliminated 1984 studies and 417 studies were eligible for full text screening. To mitigate bias, the evaluation was conducted by two impartial reviewers. In addition, the reference lists of the included research were thoroughly examined to discover any more pertinent papers that may have been overlooked. This comprehensive methodology ensured a strong compilation of studies pertinent to the diagnosis and treatment of strokes within the designated timeframe.

4.4 Result

The specified search approaches returned 3064 abstract articles. After conducting a thorough evaluation of all studies based on their title, abstract, and conclusion, we identified 125 studies related to stroke prediction, 77 studies related to stroke diagnosis, 53 studies related to stroke treatment, 141 studies related to stroke prognostication, and 21 studies related to stroke review. Out of these, 417 studies underwent a full-text review to determine their suitability, and 226 of these studies met all the criteria for inclusion in the meta analysis.

Summary: This discussion encompassed several subjects, such as the classification of journal data, the development of a methodical search approach for research pertaining to stroke, and the enhancement of technical terminology in an academic overview. We designed an extensive search methodology for the prediction and treatment of strokes spanning the years 1981 to 2023, employing databases such as PubMed, Medline, EMBASE, Web of Science, and Scopus. Furthermore, we improved the precise wording for a research paper that evaluates the impact of ML on the prevention, diagnosis, and treatment of strokes. The primary objective of the discussion was to improve the insight, technical accuracy, and organization of research methodology and summaries.

5 Data Sources and Challenges

Data types commonly addressed in cardiovascular and stroke research includes clinical tests, imaging findings, bioelectrical signals, and patient demographics. The data is collected from several sources including hospitals, research organizations, and clinical trials, frequently depending on medical technology such as electrocardiogram (ECG), MRI, and electroencephalogram (EEG) for the purpose of monitoring and diagnosing diseases. The following part highlights the types of data, the sources of data, and some of the primary challenges in collecting data and sources, in addition to suitable strategies for addressing them.

5.1 Types of Data Used

To build any model, it requires input data. For the stroke prediction model, the input data can be categorised mainly in three categories images, signal and tabular data. Each of these data has its own advantages and challenges.

Imaging Technology: Imaging technologies such as Computed Tomography (CT) and MRI play a crucial role in the diagnosis and treatment of stroke. These techniques offer refined visual representations of brain tissue and blood vessels, therefore enabling precise diagnosis and formulation of treatment strategies. Given their rapidity and effectiveness, CT scans are frequently the preferred option in emergency situations [133, 199]. Their efficacy in identifying acute hemorrhages and distinguishing between hemorrhagic and ischemic strokes is crucial for determining the suitable treatment strategy.

In contrast, MRI provides a more extensive and intricate representation of the brain [204], particularly when used alongside advanced techniques such as diffusion weighted imaging (DWI) and perfusion modeling [205]. The aforementioned techniques facilitate the timely identification of

stroke alterations in the brain that may not be visible on CT scans. MRI's ability to detect small changes in brain tissue makes it an essential instrument for precisely diagnosing stroke and assessing the magnitude of brain damage [167].

Cardiac imaging also plays a crucial role in computational stroke research as it elucidates atrial structure, remodelling, and flow dynamics that contribute to thrombus development. Imaging facilitates the characterisation of left atrial appendage (LAA) morphologies, with the Chicken Wing form linked to a significantly reduced risk of stroke or TIA, even after controlling for additional variables [52]. Furthermore, imaging has demonstrated that inflammation-induced atrial myopathy can facilitate thrombus development irrespective of atrial fibrillation [146].

MRI and CT scans are essential inputs for developing predictive diagnostic or prognostic models for stroke. These images may be assessed either by manual or automated means. Comprehensive manual inspection, albeit meticulous, is frequently time consuming and demands special skills. By contrast, automated evaluation systems have the capability to efficiently handle substantial amounts of imaging data. These automated methods have a significant potential for alleviating medical professionals' workload and improving stroke diagnosis precision and efficiency.

Signal: Electrocardiogram is generally used to monitor cardiac function of the human body. It is able to detect any abnormality in the cardiac function [130, 218]. An abnormality in the cardiac functions may increase the risk of stroke along with various other diseases. Ischemic stroke is one of the most prevalent type of stroke caused by AF [119]. ECG is the primary tool for the diagnosis of AF even when it is symptomatic [119].

The difference between human and AI based analysis of ECG is that sometimes human may miss some unusual patterns, while well-trained AI models have the potential to detect such patterns more consistently. However, like any analytical tool, AI systems are not infallible and their performance depends heavily on the quality of training data and algorithms used. There has been a lot of advancement in the ECG devices and the advent of portable ECG devices revolutionized the world. It has the capability to monitor the heart rate 24x7. This also led to the inevitable research in automatic detection of abnormal patterns in ECG so that the appropriate measures can be taken in time to prevent any serious health issues.

The signals produced by ECG can be converted into digital signals and can be used by the AI based automatic model for learning the patterns for the prediction of stroke. There are lots of work where ECG is used to predict the stroke. Sometimes ECG is used alone or sometimes ECG is combined with other health parameters for the automatic prediction of stroke [119].

Tabular Data: The application of health indicators in stroke prediction and prevention, a major cause of worldwide mortality, is of great importance. The aforementioned markers, including cholesterol, blood sugar, blood pressure, body mass index (BMI), and lifestyle elements such as smoking, diets, and physical activity, offer crucial insights into the risk of stroke. For example, hypertension is a major risk factor, while increased levels of cholesterol and blood sugar contribute to the development of atherosclerosis, which in turn raises the probability of CVD [7, 49]. Monitoring these parameters facilitates timely intervention, encompassing pharmaceutical interventions, lifestyle modifications, and other preventative measures, resulting in a decrease in stroke occurrence and improving general public health outcomes [105].

Systems that automate the analysis of tabular data inputs provide notable benefits compared to manual review, which is susceptible to errors and inefficiencies, particularly when handling extensive datasets. By utilizing artificial intelligence models trained on these specified criteria, the system is able to identify patterns and connections that may not be readily evident to healthcare professionals [182]. These algorithms have the capability to identify persons at high risk at an early stage, allowing prompt interventions and tailored treatment strategies [105, 222].

Summary: Image data provides rich, comprehensive information and may be used to leverage advanced DL approaches for developing prediction models, but it also comes with a cost in terms of processing power and preparation. While signal data is useful for capturing temporal patterns and fine grained features, noise, complexity, and large data volumes present obstacles. Tabular data is easily interpreted and structured, which makes it easier to integrate and integrate models. However, it frequently necessitates rigorous feature engineering and cautious handling of missing values. Every data type has advantages and disadvantages of its own, and the best option depends on the situation, the resources available, and the data. Combining several data types generally provides a more complete and robust solution, as it allows leveraging the complementary strengths of each modality.

5.2 Data Sources, Data Privacy and Ethical Considerations

This section provides an overview of important medical research and datasets that specifically address cardiovascular health, stroke prediction, and the application of bio-electrical signals in clinical diagnostics. The Framingham heart study (FHS) and Cardiovascular health study (CHS) are significant scientific endeavors that have significantly enhanced the knowledge of cardiovascular illness through

the use of longitudinal data collected from a large number of individuals. Contemporary datasets, including the Kaggle stroke prediction dataset, prioritize the use of predictive modeling and ML techniques to enhance the quality of stroke diagnosis and results. Furthermore, bioelectrical signal data, including EEG, ECG, and EMG, are crucial in the diagnosis of strokes, with a predilection for noninvasive methods.

Framingham Heart Study: The Framingham heart study, which began in 1948, enrolled 5,209 individuals between the ages of 30 and 62 who were residents of Framingham, Massachusetts. Participants got medical checkups every two years, which included physical examinations, blood tests, ECGs, and comprehensive medical histories [93]. In 1971, the study was expanded to incorporate the Offspring cohort, and subsequently included additional cohorts that were more varied and spanned multiple generations. Over time, modern technologies and approaches, such as enhanced imaging and genomic studies, were incorporated. The data was carefully documented, preserved, and disseminated across the scientific community, greatly enhancing the comprehension of cardiovascular disease and influencing public health policies [22, 46, 93, 99, 175, 194, 209, 212].

Kaggle: The Kaggle stroke prediction dataset is sourced from healthcare records and health surveys, encompassing data collected from hospitals and clinics. The data include medical records of patients, feedback from surveys, outcomes of clinical examinations, and findings of diagnostic testing. The dataset includes individuals who have been diagnosed with a stroke as well as those who have not [169]. It incorporates several variables such as age, gender, medical history, and health indicators. The data undergoes a process of cleaning and anonymization to guarantee privacy and adherence to legislation. The ultimate dataset has factors such as hypertension, heart disease, BMI, and smoking status. It has been prepared for public access on sites like Kaggle, where it can be used for predictive modeling and instructional objectives [51, 55, 85, 116, 140, 149, 169, 189, 209].

Cardiovascular Health Study: The CHS dataset was initiated in 1989 and includes 5,888 people who are 65 years of age or older from four different communities in the United States. Additionally, 687 African-American participants were included in order to increase the diversity of the dataset. The data gathering process involved comprehensive initial examinations and yearly followups, which consisted of physical examinations, laboratory testing, and diagnostic procedures [64]. These measures were implemented to track and assess cardiovascular well being. Important data points include clinical measurements such as blood pressure and cholesterol levels, diagnostic tests like echocardiograms

and ECGs, health assessments including medication use and diet, and tracking of cardiovascular events. Stringent data management and quality control procedures guarantee precision, and the dataset is accessible to researchers for investigations on cardiovascular health and aging, facilitating the detection of risk factors and the formulation of preventive measures [98, 118, 119, 121, 142, 144, 182].

Atrial Fibrillation Recurrence Modeling Dataset:

The AF recurrence modeling dataset provides anatomical and computational data to predict long-term AF recurrence post-ablation using 100 patient-specific biophysical models (43 paroxysmal, 41 persistent, 16 long-standing persistent). Each model incorporated variations in fibrosis, fibre orientation, electrophysiology, and ablation patterns, resulting in 1100 simulations. Simulation-derived AF metrics were combined with clinical history and imaging features to train ML classifiers. Models using all feature types achieved an AUC of 0.85 ± 0.09 , recall of 0.80 ± 0.13 , and precision of 0.74 ± 0.13 —outperforming models based on history (AUC 0.61 ± 0.14) or imaging (AUC 0.66 ± 0.17) alone. The dataset includes atrial surface meshes (VTK format) with node coordinates, triangular elements, atrial coordinates, and fibre fields [164]. Furthermore, [163] Roney et al. enables creation of a Universal Atrial Coordinate (UAC) system for standardized visualization and mapping across patient-specific atrial geometries. Collectively, these applications demonstrate the dataset's value in linking anatomical detail with translational clinical insights [146, 163].

Universities, Hospitals and Institutes: A study in Finland used a central registry to identify diabetic patients aged 45–64 requiring medication therapy [127]. It included 510 individuals from East Finland (Kuopio University Hospital) and 549 from West Finland (Turku University). Patients with insulin dependent diabetes, ketoacidosis, or serum creatinine above 200 mmol/L were excluded. The nondiabetic control group included 651 individuals in East Finland and 730 in West Finland. A separate study in Oslo (1972–1973) examined 25,915 males aged 40–49, collecting detailed stroke morbidity data from hospitals in Oslo and other areas using the ninth version of the international classification of diseases [75]. The Atherosclerosis risk in areas (ARIC) Study examined a group of 15,800 adults between the ages of 45 and 64 from four different areas in the United States. The study used B-mode real time ultrasonography to analyze data and determine the relationship between the presence of illness and the thickness of artery walls [30]. The program implemented by the university of Kuopio evaluated patients' medical records on chest discomfort, smoking habits, alcohol consumption, physical activity levels, and medication usage. The diagnoses were verified using the criteria established by the WHO, and participants were classified as hypertensive if they were taking medication

or had a high BMI [109]. The ethics committee at Hôpital Pompidou has granted approval for a study that will involve 1,715 individuals who do not show any signs of CVD. The study aims to evaluate arterial stiffness and risk factors. The study, which examined abnormal glucose levels, discovered that every patient had hypertension, with the exception of stroke and other cardiovascular conditions [107]. The study conducted by the behavioral brain sciences centre at the

Table 3 Overview of the key datasets employed in the reviewed studies related to cardiovascular and stroke research

Dataset/study name	Region	Type	Focus area(s)	Population size	Data type
ARIC Study [158]	USA	Longitudinal Cohort	Atherosclerosis, stroke, cardiovascular disease	15,792	Clinical, lab, survey
NHANES [13]	USA	Cross-sectional Survey	Nutrition, general health, chronic disease	5000/year	Survey, exam, lab
NHEFS [13]	USA	Follow-up Cohort	Mortality, cardiovascular disease, cancer	14,407	Survey, medical, mortality
Ohasama Study [26, 146]	Japan	Community Cohort	Blood pressure, stroke, cardiovascular risk	2,542	Ambulatory BP, clinical
CARDS Trial [34]	Multi-national (UK-led)	Clinical Trial	Statins, cardiovascular prevention in diabetes	2,838	Clinical trial data
CHS Dataset [60]	USA	Longitudinal Cohort	Aging, cardiovascular disease, stroke	5,888	Exam, clinical, imaging
Kaggle Stroke Dataset [67, 70–72]	Global (synthetic)	Open Dataset	Stroke prediction (ML-focused)	43,400	Tabular (demographic + clinical)
NHIRD [180]	Taiwan	Claims Database	Population health, stroke, chronic disease	798,611	Claims, prescriptions, visits

university of Birmingham recruited 25 participants who had chronic acquired brain injuries. The researchers used the Birmingham university cognitive screen (BUCS) to assess the participants' condition [24]. A study was undertaken by Meram medical school hospital with 394 patients aged 18 and above who had acute ischemic stroke (AIS), excluding those with certain health problems [192]. A separate investigation conducted at Hacettepe university neurology intensive care and stroke unit examined 205 patients with acute ischemic stroke [3]. The study utilized transthoracic echocardiography, 24 h holter monitoring, CT scans, MRI, and followup neuroimaging. The study collected EEG data from 48 stroke patients and 75 healthy persons using the Biopac MP 160 Module. Its objective was to investigate the influence of EEG recordings on the neurological health of stroke patients [84].

Snapshot of the Used Datasets: Table 3 presents a comparative analysis of several significant datasets and studies commonly employed in stroke and cardiovascular studies. The datasets originate from various domains and encompass diverse study designs, including longitudinal cohorts, clinical trials, health surveys, and electronic health records. Their differences lie in extent, demographic size, data categories, and availability.

Summary: This discussion centers on the diverse data sources employed in the context of cardiovascular and stroke research, underscoring their significance in the progression of medical research. Clinical measures and imaging outcomes, such as ECG and MRI, as well as bioelectrical signals like EEG, are among the data kinds collected from hospitals, research institutions, and clinical trials. A crucial function of these sources is to provide extensive datasets that enable researchers to investigate disease patterns, treatment results, and diagnostic instruments. In order to guarantee the accuracy and practicality of the obtained data for modern medical procedures, it is important to include a wide range of subjects and advanced technologies. Valid data sources are essential for generating influential research that enhances the treatment of patients and progresses therapeutic approaches for cardiovascular diseases and stroke. Table 3 represents an overview of the primary datasets employed in the reviewed studies, highlighting key features including types of data, population size, and focused areas.

5.3 Biomarkers in Stroke Prediction

Biomarkers are observable biological indicators, such proteins, genes, or metabolites, that might indicate abnormalities in health or disease conditions. In stroke prediction, particular biomarkers can assist in identifying individuals at elevated risk by indicating processes such as inflammation, thrombus development, or vascular damage, which are

associated with stroke. Increased concentrations of specific biomarkers, such as C-reactive protein (CRP) or particular microRNAs, have been associated with a heightened risk of stroke, facilitating early identification and preventive measures.

This comprehensive analysis assesses the efficacy of blood biomarkers in detecting ischemic stroke, with a focus on their capacity to reliably distinguish ischemic stroke from other types of stroke and non-stroke conditions. It detects multiple biomarkers, including those associated with inflammation and brain injury, that demonstrate significant diagnostic potential [208]. In a correlated investigation, scientists examined and contrasted the patterns of gene expression in individuals who had experienced ischemic stroke, those who were in good health, persons with risk factors for vascular disease, and survivors of myocardial infarction [185].

Furthermore, additional probe sets were found that effectively distinguished stroke patients from control subjects at different time intervals. The combination of these probe sets yielded a comprehensive set of 97 probes that successfully classified ischemic strokes and differentiated control groups [74]. A separate investigation evaluated the effectiveness of mean platelet volume (MPV), NT-proBNP, and hsCRP as biomarkers for ischemic stroke in patients with atrial fibrillation (AF). The results revealed that an MPV threshold of 8.85 fL had intermediate sensitivity and specificity in predicting the incidence of stroke. In addition, a prognostic model utilizing acute MRI and diverse imaging biomarkers, such as a deep convolutional neural network (CNNdeep), demonstrated superior performance compared to other approaches in forecasting the size of lesions and the prognosis of ischemic stroke patients. The CNNdeep model exhibited enhanced precision in differentiating outcomes based on the treatment approach [138]. In addition, urine biomarkers such as oxidized guanine/guanosine (OxGua) and 8-isoprostane were found to be associated with myocardial infarction, stroke, and CVD mortality in a German population. Higher levels of these biomarkers were predictive of increased risks [219]. An untargeted metabolomic profiling study successfully identified 12 plasma compounds that had a strong potential to predict ischemic stroke in hypertension individuals. This study revealed significant changes in metabolic pathways and achieved a noteworthy area under the curve for prediction [73].

Summary: The research reviewed show that blood based biomarkers and metabolomic profiling can improve ischemic stroke and CVD diagnosis, prediction, and understanding. Inflammation, neuronal damage, and metabolic dysregulation biomarkers can predict stroke incidence, mortality, outcomes, subtypes, and therapeutic responses. DL algorithms with metabolomic profiling can improve

diagnostic accuracy and customized treatment options in acute ischemic stroke and hypertension patients. These findings highlight the need to validate and integrate biomarkers into clinical practice to improve ischemic stroke and CVD patient care and outcomes.

5.4 Challenges in Data Acquisition and Usage

Several data related issues impact the quality and reliability of findings in cardiovascular and stroke research. Maintaining uniformity across various cohorts, time periods, and measuring procedures is a significant challenge for large scale investigations such as the FHS, making data accuracy and consistency crucial concerns [175]. To reduce inconsistencies, it is advisable to establish consistent procedures, conduct regular audits, and provide training for data collectors. Missing or partial data is an additional obstacle, especially in datasets like the Kaggle Stroke Prediction Dataset, where not all participants undergo identical tests. Imputation techniques and complex statistical approaches can address this challenge, while more thorough data collection at the source can minimize gaps. Precision in categorization, such as identifying stroke cases with clinical and imaging data, is crucial, and multi level verification by experts helps avoid mischaracterization. Ethical concerns regarding patient privacy and data security can be managed through de-identification, encryption, and strict compliance with rules, safeguarding patient rights.

It is also important to recognize the distinction between routinely collected data (such as electronic health records, registries, and insurance claims) and data gathered under structured research protocols (such as clinical trials and longitudinal cohort studies). Routinely collected data are often large-scale and cost-effective but present challenges including inconsistent coding, incomplete records, and variability across institutions and time periods. In contrast, protocol-driven data collection ensures greater standardization, reliability, and quality control but may be limited by smaller sample sizes, narrower populations, and higher costs. Both approaches have complementary strengths, and careful integration of these data sources can substantially improve the robustness and generalizability of findings in stroke research.

Additionally, the extensive range and variety of data sources provide complications. The lack of diversity in many datasets, such as those from earlier cardiovascular tests, restricts the generalizability of scientific conclusions. Expanding the cohorts of the CHS, which exhibit increased representation from diverse ethnic and socioeconomic groups, serves to enhance the generalizability of research findings [64]. The presence of outdated data is a matter of concern, particularly when previous research fail

to accurately represent contemporary diagnostic methods or medical procedures. Continuously updating datasets with the most recent clinical and technical breakthroughs, such as advanced imaging or genetic analysis, guarantees their relevance in current research. By effectively tackling these obstacles by implementing more advanced methods of data gathering, manipulation, and ethical standards, researchers can elevate the standard and impact of studies, therefore resulting in more advanced diagnostic tools and treatments for cardiovascular diseases and stroke.

5.5 Healthcare Data Preprocessing Issues

Data preprocessing is an essential approach in medical data analysis due to the intrinsic complexity, heterogeneity, and variability of medical data obtained from many sources. It tackles vital concerns including absent values, redundancies, inconsistencies, and noise, which are crucial to rectify for guaranteeing precise study results, efficient clinical decision-making, superior model performance, and heightened patient safety [78]. Hassler et al. applied stringent data preparation to the Toledo Study for Healthy Ageing dataset of 474 seniors with 284 variables, many of which had missing or non-standardized. The pipeline included data cleansing, removal of insignificant features, expert-assisted outlier management, and creation of variables like BMI and mobility ratings. Missing data were addressed with a customised multiple imputation method using random forests (CALIBERrfimpute), accounting for MAR and MNAR to ensure methodological rigour.

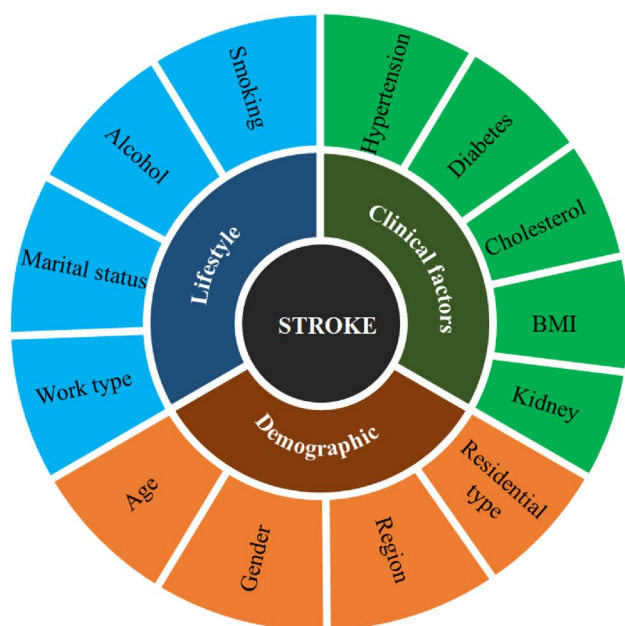


Fig. 4 A comprehensive wheel of stroke risk factors: demographics, lifestyle choices, and clinical conditions

Mavrogiorgos et al. demonstrated the necessity for domain-specific preprocessing procedures by proposing a multi-tiered healthcare data cleansing platform that includes modules like a Gateway, Data Storage, and Cleaning Actions. This approach employed Python for data standardisation, MongoDB for storage, and KNN for imputing missing values, while methodically tackling data duplication and outliers. The validation and verification phases confirmed consistency with actual clinical datasets, such as those from Kaggle, underscoring the significance of organised, automated data pipelines in health informatics [125]. Khasawneh et al. examined preprocessing in a qualitative educational context, emphasising the identification and alleviation of cultural, linguistic, and gender biases via thematic analysis of questionnaires and focus groups. Their work highlights the interpretative and equity-focused aspects of data analysis in the social sciences [97].

These studies show that preprocessing improves data validity and reliability but must be tailored to domain goals such as accuracy in healthcare versus fairness in education. Hybrid methods combining quantitative rigor with qualitative insight hold promise. In healthcare, domain-specific ontologies can improve imputations and anomaly detection, while in education, NLP can help detect cultural biases. Adaptable frameworks could enhance quality and ethical standards of data-driven research across fields.

6 Key Risk Factors for Stroke

Stroke is usually caused either by the bleeding in the brain or due to the blockage of blood supply to the brain. For the diagnosis, detection, prediction and prognosis of the stroke, various risk factors are analysed and studied in details. A comprehensive diagram depicting the diverse risk factors associated with stroke, categorized into three primary categories: Demographic, Lifestyle, and Clinical factors are presented in Fig. 4. This detailed visual depiction clearly illustrates how several factors, including personal demographics, lifestyle choices, and clinical conditions, can influence stroke risk.

A risk factor is either an internal or external or both factors (characteristics, condition or behaviours) that increases the chances of occurrences of stroke. Researchers have developed various automatic or semiautomatic tools and models for diagnosis, detection prediction and prognosis of cardiovascular disease. The models take risk factors as input and then produces desired output by analysing the input. Study of the risk factors becomes more important when a risk factor is changeable. Not all the risk factors are changeable (age, gender) but some of them can be changed or controlled. Studies have found that by controlling the

few risk factors, the chances of occurrence of stroke can be reduced [169]. While studying more than 200 related research papers, we came across more than 50 risk factors. When we analysed closely, we found that some of these risk factors are very important and analysed by most of the works.

6.1 Demographic Factors

The influence of demographic risk factors plays an important role in establishing the probability of an individual experiencing a stroke. Age, gender, ethnicity, and socioeconomic level are well established variables that are consistently associated with differences in stroke occurrence and health consequences [118, 166].

Age: Age is a non-controllable risk factor for stroke but it is one of the most important factors. Studies have shown that the older you are the more vulnerable you are to the stroke [63, 135]. Although, adolescents may also experience stroke but it is more likely because of the blood pressure, obesity and diabetes in young generation. Out of all the previous studies that we have analysed, 90% of them used age as one of the risk factors either for diagnosis, prediction or prognosis of stroke [88, 161].

Gender: Studies have shown that the women are more vulnerable of a stroke than the men. One of the primary reasons is that the women live longer than the men therefore more vulnerable of stroke. The ratio of stroke in men is one 1:6 and in women is 1:5. The longer life expectancy of women leads to higher rate of occurrences of stroke in women. Therefore, the risk factor gender is closely related with the age [30].

Region: Geographical location is a major risk factor for stroke, with differences in stroke mortality and incidence rates noted worldwide. The risk of stroke is influenced by a variety of factors that vary by location, including socioeconomic level, access to healthcare, dietary habits, and the incidence of diabetes and hypertension. Furthermore, cultural dietary practices that increase blood pressure and stroke risk include increased salt intake in some Asian countries. Reducing the worldwide burden of stroke requires addressing regional differences through focused public health measures, expanding access to healthcare, and encouraging healthy lifestyles [60].

Residential Type: The diagnosis and prediction of stroke risk can be greatly influenced by the kind of living area—rural, suburban, or urban [150]. Urban regions frequently provide easier access to medical facilities, specialists, and cutting edge diagnostic equipment, which makes it easier to identify and treat stroke risk factors like diabetes and hypertension early on. On the other hand, there may be difficulties in rural locations, such as slower emergency response times,

fewer medical personnel, and restricted access to healthcare services, which might delay the diagnosis and treatment of strokes [126].

6.2 Clinical Factors

Analyzing clinical risk factors is crucial for understanding stroke risk and formulating efficient prevention measures. Medical disorders such as hypertension, diabetes, and excessive cholesterol greatly enhance the probability of experiencing a stroke [34, 71, 119, 195].

Hypertension: It is a situation when the pressure in the blood vessels is very high. It does not show any symptoms but it may lead to stroke. It is a controllable risk factor, but it is most likely to occur with a lifetime probability of 90% in the age group of 80 years and above [49, 129, 198].

Diabetes: When the glucose (sugar) in the blood vessels is high and if not treated, it leads to serious health problems over the time. It is a well established risk factor for the diagnosis and prognosis of the stroke [71, 156]. Diabetes can directly affect the cerebral vessels which may lead to stroke [71].

Cholesterol: Cholesterol is a very useful and waxy substance needed to build healthy cells may also lead to stroke. A high level of cholesterol can develop fatty deposits in the blood vessels which may eventually lead to the blockage in the blood vessels and therefore high chances of stroke [41]. Cholesterol is a controllable risk factor which can be controlled by following a healthy lifestyle, healthy diet, proper exercise and avoiding stress etc. [200].

Body Mass Index: BMI indicates the amount of fat in the body. Several studies have shown that the association of stroke with high BMI. The value of BMI is not identical in different genders. Therefore, BMI has to be taken into consideration with the gender [104]. It is a controllable risk factor and with healthy diet, proper exercise and healthy lifestyle, therefore reducing the overall effect of the occurrences of the stroke, hypertension, diabetes, and atrial fibrillation.

Kidney: Given that chronic kidney disease (CKD) considerably raises the risk of stroke, renal health is essential for both predicting and diagnosing strokes [45]. The kidneys are essential for blood pressure regulation and electrolyte balance maintenance, two processes that are critical for cardiovascular health. Additionally, patients with CKD are more likely to have diabetes and cardiovascular illnesses, which increases their risk of stroke [143].

6.3 Lifestyle Factors

Lifestyle factors play an important effect on an individual's susceptibility to stroke and are of paramount importance in

the prevention of this condition. Certain behaviors, including smoking, excessive alcohol use, poor diet, and physical inactivity, significantly increase the risk of stroke. A comprehensive grasp and effective management of these changeable lifestyle elements are crucial for both personal and societal health initiatives targeting the reduction of stroke occurrence.

Smoking: The smoking is directly associated with heart rate and blood pressure. Whenever a person smokes cigarette, it instantly increases the heart rate and blood pressure of that person [179]. Along with this cigarette smoking is also one of the risk factors for the development of atherosclerosis. Development of these factors are proportional in all the gender [75, 176, 202]. Our analysis of previous works shows that more than 60% works considered smoking as a risk factor and utilized it as an important attribute for the automatic diagnosis, prediction and prognosis model.

Alcohol: Drinking alcohol increases the risk of stroke significantly, affecting both hemorrhagic and ischemic forms [26]. Excessive alcohol use can result in hypertension, a significant risk factor for stroke. Furthermore, the frequent heart rhythm condition known as alcohol induced atrial fibrillation raises the risk of blood clots developing, moving to the brain, and resulting in ischemic stroke [54]. Additionally, long term alcohol consumption weakens blood arteries, increasing the risk of their rupture and the potential for hemorrhagic stroke [159].

Marital Status: Given that it can affect a stroke's risk and consequences, marital status is an interesting factor in stroke diagnosis and prediction. Studies show that married people frequently have better recovery results and a decreased risk of stroke than single people [180]. The presence of a spouse can guarantee adherence to medical treatments, promote healthier lifestyles, and offer emotional support, all of which are critical in controlling stroke risk factors like high blood pressure, diabetes, and cholesterol [191]. On the other hand, those who are widowed, divorced, or single may not have these support networks, which could lead to increased stress, unhealthy habits, and delays in getting care, all of which raise the risk and intensity of stroke [213].

Work Type: Because job demands and occupational environments have a direct impact on lifestyle and health habits, the sort of work a person does can have a major impact on stroke risk prediction and diagnosis. Occupying sedentary employment, which are prevalent in office environments, raises the risk of stroke by promoting obesity, cardiovascular issues, and physical inactivity. High stress jobs can cause long term stress and hypertension, which raises the risk of stroke even more [193]. In contrast, professions involving intense physical labor may result in higher physical stress and increased risk of hypertension and heart disease. Work schedules that interfere with sleep patterns and

circadian rhythms, such shift work, can also have a negative impact on cardiovascular health [50].

Summary: Stroke risk is determined by a confluence of demographic, clinical, and lifestyle parameters. The demographic factors of age, gender, and ethnicity have a substantial impact on the susceptibility to stroke, with older persons and specific ethnic groups experience elevated risks. Pre-existing health conditions and family history are clinically associated with increased susceptibility, but lifestyle choices like nutrition, physical exercise, smoking, and alcohol use represent modifiable risk factors. The implementation of focused interventions and preventive measures is essential in decreasing the overall occurrence of stroke and enhancing long term health results among different demographic categories.

7 Analytical Tools and Techniques

A wide range of tools and techniques, including both conventional scoring systems and sophisticated ML and DL models, are employed in stroke prediction and management. Clinical scoring based risk assessment methods employ many parameters like age, medical history, and medications to approximate an individual's sensitivity to stroke. Analysis of correlations between risk factors and outcomes using statistical approaches offers a more empirically based approach to predicting strokes [96]. By analyzing vast, intricate datasets and revealing patterns that may not be apparent using conventional approaches, ML techniques improve prediction accuracy. Advanced DL models enhance this power by integrating more complex algorithms to interpret multidimensional data, providing more personalised and accurate risk evaluations. Collectively, these tools and strategies encompass a wide range of methods designed to enhance the prognosis of strokes and the result for patients.

7.1 Risk Scores and Clinical Tools

Stroke risk scores are essential tools that allow medical professionals to assess an individual's likelihood of having a stroke within a specific timeframe. These scoring methods use several risk factors, such as age, gender, blood pressure, diabetes status, and smoking behaviors, to provide a thorough evaluation of the risk of stroke [65, 120]. This discussion will explore different types of stroke risk scores, explaining their components and clarifying their therapeutic uses.

Traditional techniques for stroke analysis involve a wide range of tools and assessments, such as risk scores and clinical evaluations, to accurately diagnose and treat stroke. The national institutes of health stroke scale (NIHSS) is a

fundamental tool used to measure the severity of strokes, evaluate neurological impairments, and inform treatment choices [68, 211]. In addition, risk scores such as the ABCD² score assist in forecasting the likelihood of a stroke following a transient ischemic attack (TIA), facilitating prompt intervention and preventive actions [178]. The modified ranking scale (mRS) offers a standardized approach to assess functional outcomes and impairment following a stroke, providing valuable information for rehabilitation techniques and patient care [187]. Furthermore, the Alberta stroke program early CT score (ASPECTS) is employed to promptly evaluate ischemia alterations on brain CT scans, providing guidance for treatment choices and forecasting patient outcomes [18]. By incorporating conventional techniques such as risk scores, clinical assessments like NIHSS and mRS, and imaging evaluations such as ASPECTS, a thorough and all encompassing approach to stroke analysis can be achieved. By effectively combining these techniques, healthcare providers can maximize stroke therapy, enhance patient outcomes, and decrease the impact of stroke related impairment.

National Institutes of Health Stroke Scale: The NIHSS is a standardized evaluation tool used to evaluate the severity of stroke symptoms in individuals [68]. While the NIHSS is not specifically intended for predicting stroke occurrence, it is extremely important in the management and prognosis of strokes. The NIHSS assists healthcare providers with making intelligent decisions regarding treatment, evaluating responsiveness to therapies, and predicting outcomes after a stroke by thoroughly testing neurological functions including consciousness, vision, motor abilities, coordination, and language [211]. A higher NIHSS score upon presentation is usually associated with a more severe stroke and indicates a worse prognosis. Although the NIHSS does not directly predict stroke episodes, it is extremely important in directing therapeutic management and predicting outcomes for people who have had a stroke [11, 165, 183].

A study of fifty individuals with acute ischaemic stroke used the NIHSS to monitor recovery after treatment. Neurological advancement was defined as a decrease of 4 or more points, showing enhancement within 7 to 10 days and later. Patients with embolic strokes in the middle cerebral artery had comparable recovery, while lacunar infarcts showed modest improvement. Initial NIHSS scores did not consistently predict recovery, indicating stroke subtype affects predictive significance [211]. The Kansas City Stroke Study prospectively examined 184 persons [106], comparing OPS to NIHSS with assessments at enrollment, 1, 3, and 6 months. Both OPS and NIHSS correlated with the Barthel ADL index and SF-36 PFI, explaining variance in these measures, but OPS showed slightly better predictive efficacy, especially for mild and moderate strokes.

A study from the Leuven Stroke Genetics Study investigated whether including changes in NIHSS within the initial 24 h could enhance the precision of predicting functional outcomes at three months. The incorporation of the rapid decrease in NIHSS (RR NIHSS) with baseline NIHSS, age, and ischaemic heart disease in a multivariate model substantially enhanced prediction accuracy for desirable functional outcomes. This indicates that early NIHSS alterations are vital predictors of long term prognosis and have to be incorporated into prediction models to enhance outcome assessments [216].

In summary, the NIHSS is a reliable instrument for evaluating stroke severity and forecasting diverse outcomes; nevertheless, its predictive efficacy may fluctuate based on stroke subtype and assessment timing. Research recommends for the incorporation of initial alterations in NIHSS scores and the evaluation of patient specific variables such as age and comorbidities to enhance predictive models.

Orpington Prognostic Scale: The OPS is a clinical tool that evaluates stroke severity and predicts functional outcomes, including rehabilitation needs. Derived from the Edinburgh Prognostic Score, it incorporates motor deficit, proprioception, balance, and cognitive function, enhancing predictive capacity for long-term recovery [56, 90]. The exam is more thorough with a cognitive test, especially helpful for older patients with cognitive problems affecting recovery. OPS is a weighted rating system ranging from 1.6 (best prognosis) to 6.8 (worst prognosis), assessing motor performance, proprioception, balance, and cognition. Higher scores indicate greater disability and lower probability of autonomous recovery.

Numerous studies have corroborated the OPS's prognostic precision regarding stroke rehabilitation outcomes, particularly in elderly individuals [56, 210]. Kalra and Crome established that OPS, administered two weeks after a stroke, is strongly associated with functional recovery and the capacity for patient discharge within several weeks. Specifically, OPS demonstrated enhanced prognostic value in individuals with dementia, where cognitive deterioration frequently hinders rehabilitation [90].

Studies show OPS predicts stroke outcome, especially in older people, strongly linked to functional status at release. Kalra et al. noted OPS at two weeks had robust predictive value for identifying patients who benefit from intense rehabilitation. OPS below 3.2 led to discharge after three weeks, while values beyond 5.2 required long term care, essential for resource allocation in stroke management [91]. While Whitall et al. showed that although OPS is a reliable predictor of early recovery, it may also be associated with lingering deficits, particularly in motor function and daily tasks, that continue after patients achieve independence [207].

The OPS is an essential instrument for evaluating stroke severity and forecasting rehabilitation requirements, especially in geriatric patients. The comprehensive methodology, integrating motor, sensory, and cognitive evaluations, offers medical professionals an effective framework for decision making. Research regularly confirms its predictive precision, especially in distinguishing between patients who will gain from rehabilitation and those necessitating long term care. As stroke management advances, the OPS continues to be an essential tool for customizing therapies to meet individual patient requirements, hence enhancing recovery results and optimizing healthcare resource distribution [56, 90, 91, 207, 210].

Alberta Stroke Program Early CT Score (ASPECTS): The ASPECTS offers a standardized approach to assess early ischemia alterations on non-contrast CT scans in individuals with acute ischemic stroke. The ASPECTS system divides the territory of the middle cerebral artery into ten sections and evaluates each region for signs of ischemia. It then assigns a score ranging from 0 to 10 based on the findings. Greater scores suggest a lower degree of ischemia damage and are linked to more favorable treatment responses and outcomes. The ASPECTS is extensively utilized in both clinical practice and research to assist in making treatment decisions, including thrombolytic therapy and endovascular treatments, and to offer prognostic information in the management of acute ischemic stroke [151].

A study evaluated the predictive ability of ASPECTS in hyperacute ischemic stroke patients treated with thrombolytic medication. A total of 203 individuals received intravenous alteplase within 3 h at two North American teaching hospitals. CT scans before therapy were evaluated with ASPECTS. Main outcomes were symptomatic intracerebral hemorrhage and functional result at 3 months. Results showed 75% of anterior circulation ischemia patients had ischemic alterations on initial CT scans. Baseline ASPECTS inversely correlated with stroke severity by NIHSS and predicted both functional outcome and symptomatic intracerebral hemorrhage. ASPECTS showed notable sensitivity and specificity for these outcomes, indicating it is an uncomplicated and dependable method to identify patients unlikely to achieve independent recovery with thrombolytic treatment [18].

Anticoagulation and Risk Factors in Atrial Fibrillation (ATRIA): The ATRIA score is a risk assessment measure utilized to evaluate the likelihood of stroke and bleeding in individuals with atrial fibrillation (AF) who are contemplating anticoagulant medication. The ATRIA score is derived from a comprehensive study involving a large community based group. It considers many clinical indicators to determine an individual's likelihood of having a stroke and enduring bleeding problems. The ATRIA score

comprises several factors, namely age, hypertension history, diabetes history, congestive heart failure history, stroke/TIA history, bleeding history, and renal function (estimated glomerular filtration rate). Every component is allocated a distinct weight or score according to its proportionate impact on the risk of stroke and bleeding [67].

The ATRIA Study aims to assess the prevalence of atrial fibrillation (AF) and predict the number of AF patients in the United States through 2050. An review of persons diagnosed with atrial fibrillation in California from 1996 to 1997 revealed 17,974 occurrences, with 45% occurring in individuals aged 75 and older. The overall prevalence rate was 0.95%, signifying a significant cause of atrial fibrillation, projected to escalate 2.5 fold by 2050. This discovery highlights the considerable influence of atrial fibrillation on the elderly and its ramifications for the management of cardiac rhythm disorders and stroke prevention at a national level. The research offers essential insights for healthcare policies and initiatives focused on enhancing atrial fibrillation management and decreasing related morbidity and death [67].

A novel stroke prediction model for atrial fibrillation patients was developed using the ATRIA cohort and validated with the ATRIA CVRN cohort. It identified eight predictors: age, stroke history, female gender, diabetes mellitus, heart failure, hypertension, proteinuria, and renal dysfunction, each scored by weighted coefficients. The ATRIA stroke risk score showed better predictive ability than existing tools, with higher C-index values and improved net reclassification, enhancing stroke risk prediction [181].

ABCD²: The ABCD² score is a clinical tool utilized for predicting the likelihood of a stroke occurring after a TIA, sometimes known as a “mini-stroke”. It assists healthcare workers in identifying individuals who may have an increased likelihood of developing another stroke and aids in making decisions regarding additional diagnostic tests and treatment [154].

Multiple population-based and longitudinal cohort studies have evaluated the effectiveness of imaging markers and clinical scores in predicting stroke risk following TIA. Research indicates that vascular imaging, especially regarding carotid stenosis, serves as a robust independent predictor of both early and long-term stroke recurrence, with risk escalating in accordance with the severity of the condition [101, 154, 178]. Clinical scores like ABCD2 exhibited limited predictive value in neurologist-verified TIA cases, although they proved more beneficial in potential TIA instances evaluated by non-specialists [178]. Enhanced models like ABCD3 and ABCD3-I demonstrated superior short-term predictive accuracy compared to ABCD2. Additionally, the incorporation of vascular imaging, including carotid and intracranial artery stenosis, significantly enhanced long-term risk stratification [101]. These studies

indicate that the prediction of stroke risk following TIA is enhanced by the integration of vascular imaging with clinical evaluation [101, 154, 178].

Modified Rankin Scale: The mRS is a widely employed tool for evaluating disability or dependence in patients after stroke or cerebral injury. It offers a thorough assessment of functional outcomes worldwide and is commonly used in clinical practice, medical studies, and research. The score is obtained by interview with the patient or caregiver about capacity for everyday tasks and needed assistance. It plays a vital role in classifying disability and guiding treatment and rehabilitation, providing essential information about overall functional outcome [16].

Research initiatives have aimed to create and provide tools that can precisely forecast functional outcomes and mortality prior to an ischaemic stroke. The Bologna Outcome Algorithm for Stroke (BOAS) is a straightforward yet accurate prognostic scale that operates without the need for computational assistance. In a cohort of 221 ischaemic stroke patients and 100 validation cases, the BOAS scale found five binary risk factors substantially correlated with a mRS score exceeding 2 at nine months post-stroke. The risk factors were a NIHSS score of ≥ 10 , age ≥ 78 years, necessity for a urinary catheter, need for oxygen therapy, and the presence of upper limb paralysis at discharge. The existence of two or more risk variables markedly elevated the likelihood of $mRS > 2$ (91%) and mortality (42%), while the absence or presence of only one risk factor diminished the risk to 24% and 2%, respectively. The BOAS scale had strong predictive efficacy, evidenced by an area under the ROC curve (AUC) of 0.891 and an overall prediction accuracy of 83.9%. The validation group confirmed these findings, demonstrating comparable accuracy (79.0%) and an AUC of 0.839, indicating that the BOAS scale is a dependable instrument for forecasting long term dependence or death based on readily available clinical factors [132]. Simultaneously, ML methods like tree boosting and MLPs were compared with logistic regression for predicting 90-day post-stroke mRS. ML improved accuracy, but understanding clinical predictors remained crucial for interpretability and use [225].

CHADS₂/CHA₂DS₂-VASc: The CHA₂DS₂-VASc score is a clinical prognostic tool utilized to assess the likelihood of stroke in individuals with non-valvular AF. The scoring system allocates points according to prevalent risk factors for stroke, such as congestive heart failure, hypertension, age, diabetes mellitus, past stroke or TIA, vascular disease, and sex category. The aggregate point total is utilized to calculate the risk of stroke during a specified time period, typically one year, with a point value ascribed to each factor. As the CHA₂DS₂-VASc score increases, the likelihood of experiencing a stroke also increases [20, 95, 128].

Instead of the CHADS₂ score, which is usually used to figure out a person with atrial fibrillation's risk of having a stroke, Letham et al., tries to create accurate predictive models that can be understood by experts. The authors simplify intricate multivariate feature spaces into assertions that are straightforward to interpret by utilizing decision lists: if...then... statements. Using UCI ML repository data, they introduce the Bayesian list machine, a generative model that generates a posterior distribution over decision lists and promotes sparsity using a unique structure. The Bayesian list machine has an AUC equivalent to the widely used CHADS₂ score, indicating it may be a more accurate and interpretable stroke risk assessment tool in clinical practice [111].

Inspired by developments in customized medicine, Friedman et al., work presents Bayesian Rule Lists as a CHADS₂ score alternative. The approach discretizes complicated, multivariate feature spaces into straightforward decision statements using decision lists, which are made up of if...then... statements. Using the MarketScan Medicaid Multi-State Database (MDCD), which contains administrative claims data for 11.1 million Medicaid members from various states, we developed the stroke prediction model [110].

Framingham Stroke Risk Score (FSRS): The FSRS is a vital tool used to evaluate the likelihood of experiencing a stroke, derived from the influential FHS. The tool calculates the likelihood of an individual experiencing a stroke within a specific timeframe by taking into account various risk variables including age, gender, blood pressure, diabetes, smoking history, and cardiovascular disease. Healthcare practitioners use the FSRS to identify people at high risk and customize preventative treatments, including lifestyle changes and medication. Although the FSRS provides useful insights into risk stratification [93, 175].

A separate study assessed the predictive value of coronary artery calcification (CAC) in anticipating stroke risk beyond traditional Framingham Risk Score (FRS) components. Analysis of data from 4,180 participants in the Heinz Nixdorf Recall Study revealed that baseline CAC levels strongly forecasted stroke occurrences, especially in persons under 65 years of age, regardless of atrial fibrillation status. Moreover, CAC offered a more nuanced risk stratification for individuals deemed low to moderate risk by FRS, indicating that the inclusion of CAC measurements could improve stroke risk prediction in these cohorts and potentially inform more focused preventive measures [81].

Alongside conventional scoring systems, mobile health technologies such as the Stroke RiskometerTM have been created to assess stroke risk. The application was validated with data from 9,501 individuals in New Zealand, Russia, and the Netherlands. In comparison to existing stroke prediction algorithms like the FSRS and QStroke, the Stroke

Riskometer™ demonstrated performance on par with FSRS and surpassed QStroke. Nonetheless, despite its portability and simplicity, the overall discriminatory capability was constrained, with C-statistics and D-statistics varying from 0.51 to 0.56 and 0.01 to 0.12, respectively, signifying that additional refinement is necessary to improve accuracy and applicability across diverse populations [148].

Acute Stroke Registry and Analysis of Lausanne (ASTRAL): The ASTRAL score is a useful measure for predicting the functional prognosis and death of patients with acute ischemic stroke. Based on an extensive stroke registry, it includes important clinical and demographic factors such as age, severity of stroke, functional level before the stroke, and other conditions like atrial fibrillation, diabetes, and hypertension. Healthcare providers can use the ASTRAL score to determine the risk levels for poor functional outcome or mortality in patients who have had an acute ischemic stroke [139].

ML algorithms have been progressively investigated for predicting stroke outcomes. A comparative research of three ML models (DNN, RF, and LR) utilizing the ASTRAL score revealed the capacity of DL to improve long term outcome prediction in patients with acute ischaemic stroke. In a cohort of 2604 people, the deep neural network model demonstrated a markedly elevated AUC (0.888 compared to 0.839; $P < 0.0001$), signifying enhanced efficacy in forecasting favourable outcomes. Nonetheless, the RF and logistic regression models did not significantly surpass ASTRAL when employing the identical six variables, indicating that DL techniques may provide distinct benefits in managing intricate data patterns and enhancing the overall precision of stroke prognostication [80].

Summary: A great deal of research has advanced our ability to forecast and comprehend the risks and consequences of stroke. Several clinical scoring systems, including the NIHSS, and ATRIA stroke risk score, have been assessed for their effectiveness in predicting the severity of strokes and the resulting functional outcomes. Furthermore, ML methods such as decision lists and Bayesian Rule Lists have developed as effective tools for creating precise and understandable predictive models. In addition, the incorporation of genetic data, such as polygenic risk scores, and imaging techniques, such as coronary artery calcification, has yielded valuable insights in enhancing stroke risk evaluation beyond conventional clinical risk factors. These findings highlight the significance of using diverse data and sophisticated analytical methods to improve the prediction of stroke risk and support focused preventative initiatives. This eventually leads to better patient outcomes and helps guide clinical decision making in stroke management.

7.2 Statistical Analysis Models

Statistical approaches play a crucial role in stroke research, providing robust techniques for assessing data and generating significant insights. The Chi-square test, Cox regression, multivariate logistic regression, and Kaplan-Meier analysis are among the most effective techniques for examining various facets of stroke incidence, progression, and outcomes.

Cox Proportional Hazards Model: The Cox proportional hazards model is a semi-parametric technique employed in survival analysis to investigate the impact of different factors on the duration until an event, such as mortality or the recurrence of a disease. The assumption is that the ratio of dangers across persons remains constant over time, hence the concept of “proportional hazards”. The model calculates the comparative risk of the occurrence for several groups without presupposing any assumptions about the fundamental distribution of survival duration [113].

In a comprehensive analysis of cardiovascular risk factors, Simone et al., employed the Cox proportional hazard model to examine the relationship between the stroke volume/pulse pressure ratio and cardiovascular risk in adults with arterial hypertension. Data from electronic health records of patients at a hypertension clinic were analysed. The stroke volume to pulse pressure ratio was analysed, and cardiovascular risk was evaluated utilising ML techniques. Upon controlling for confounding variables, a larger stroke volume/pulse pressure ratio was strongly correlated with elevated cardiovascular risk, regardless of recognised risk factors [49].

Similarly, a separate study employed Cox proportional hazard analysis to investigate whether depression symptoms could forecast future stroke occurrence. Data from NHANES I and NHEFS, encompassing a cohort of 6,095 persons aged 25 to 74 with no prior history of stroke, was employed, with patients monitored for a duration of up to 22 years. The analysis demonstrated that baseline depressed symptoms were consistent predictors of stroke incidence in age adjusted models for all demographic classes [87].

The Ohasama Study in Japan examined blood pressure and stroke risk in adults without prior stroke, showing that 24-hour, daytime, and nighttime ambulatory blood pressures predicted stroke better than screening values, with daytime readings more predictive than nighttime [141]. A subsequent cohort of 1766 adults followed for 11 years found morning home blood pressure more strongly associated with stroke risk than evening values, underscoring the predictive value of morning measurements [8].

In a separate analysis, the Collaborative Atorvastatin Diabetes Study (CARDS) trial showed that atorvastatin significantly reduced stroke risk and incidence in diabetic patients compared to placebo [82], while a long-term study of 1,715

hypertensive patients found that increased carotid-femoral pulse wave velocity was linked to a 1.72-fold higher risk of stroke mortality, highlighting aortic stiffness as a valuable prognostic marker [107].

Finally, the Heinz Nixdorf Recall research examined the prognostic significance of CAC for stroke occurrences in 4180 healthy individuals. CAC was examined as a stroke predictor along with vascular risk variables using Cox proportional hazards regression analysis. Stroke patients had considerably higher baseline CAC readings than non-stroke patients. After controlling for age, systolic blood pressure, and smoking status, $\log_{10}(CAC + 1)$ predicted stroke with a hazards ratio of 1.52 in a multivariable Cox regression model [81].

Logistic Regression Models: Logistic regression is a commonly employed statistical technique for binary classification problems, in which the result is a binary value of either 0 or 1 (i.e., disease or absence of disease). This method employs a logistic function to represent the likelihood of the dependent variable and calculates the associations between the binary result and one or more predictor variables. In contrast to linear regression, logistic regression is employed when the response variable is of a categorical form [188]. Logistic regression, Within the statistical framework, functions as an instrument for inference, enabling researchers to quantify relationships between predictor variables and a result, evaluate the significance of effects, and assess model calibration. The emphasis is on elucidating correlations and comprehending the underlying mechanisms inside the data rather than only maximizing the predictability.

Johnston et al. investigated the association of infarct size, six clinical variables, and three-month outcomes in ischemic stroke patients using RANTTAS data, incorporating 256 patients. Multivariable logistic regression with bootstrap validation demonstrated strong discriminative ability, with ROC areas between 0.79 and 0.88, and calibration curves confirmed robustness, showing near perfect agreement between observed and predicted outcomes. Combining clinical and imaging factors significantly improved prediction accuracy compared to using either alone for forecasting recovery, mortality, or severe impairment [86]. In a complementary study using MR DWI and clinical factors, 66 patients from a North American hospital were analyzed, with validation on 63 Australian patients. Logistic regression with NIHSS score, stroke onset to MR DWI, and ischemic brain tissue volume achieved 0.77 sensitivity and 0.88 specificity, showing these variables predicted recovery most effectively [13].

Further, prediction models for in-hospital mortality after ischemic stroke were developed using Get With the Guidelines—Stroke Program data from 274,988 patients across 1,036 hospitals. Logistic regression identified independent

mortality predictors, with separate models for patients with and without NIHSS scores. Mortality was 5.5%, and discrimination was good, with C statistics of 0.72 in validation, 0.85 with NIHSS, and 0.83 from NIHSS alone, while calibration was excellent [183]. Another study predicting one-year ischemic stroke recurrence in 332 first-stroke patients applied logistic regression on clinical risk factors and retinal features. The integrated model identified cerebral atherosclerosis, white matter lesions, heart illness, hemorrhage, and exudate as independent risk factors, achieving 72.5% sensitivity and 70.7% specificity [223].

Kaplan–Meier Analysis: Kaplan–Meier Analysis is a non-parametric statistical measure employed for the estimation of the survival function based on time-to-event data. It generates a step function that quantifies the likelihood of survival beyond specific time intervals. Kaplan–Meier curves are frequently utilized for the purpose of comparing survival rates among various groups, while the log-rank test is widely used to ascertain statistical significance [160].

Asymptomatic cerebral emboli (ES) detected by Doppler ultrasound were evaluated for prognostic value in carotid artery stenosis patients, with ES present in 36.9% of 111 individuals; symptomatic patients showed an inverse relationship between ES frequency and time since last symptoms, plaque ulceration increased embolic stroke risk, and Kaplan–Meier analysis confirmed that baseline ES raised the probability of TIA and stroke over time [129]. In a multicenter prospective cohort of 202 individuals with a mean 34-month follow-up, Kaplan–Meier analysis showed an annual incidence of 3.1% ipsilateral stroke/TIA, and although transcranial Doppler identified ES, they did not increase ipsilateral risk, nor significantly affect survival free of ipsilateral events, though they may serve as a marker for cerebrovascular risk [2].

Chi-Square Analysis: Chi-square analysis is a statistical procedure employed to ascertain the presence of a statistically significant relationship between two category variables. This analysis compares the observed frequencies within each category with the anticipated frequencies in the absence of any correlation between the variables. This test computes a chi-square statistic, which, in conjunction with degrees of freedom, is employed to evaluate if the observed disparities are attributable to random variation. It finds widespread use in goodness of fit tests, contingency tables, and independence testing within disciplines such as humanities and sciences [172].

In a case series examining the correlation between clinical features and aspiration occurrence within five days post-acute stroke, researchers utilized Chi-square analysis to determine significant connections. Out of a group of 55 stroke patients, 38% of them had aspiration, and in 67% of those cases, the aspiration was quiet. The research

uncovered notable connections between dysphonia, dysarthria, aberrant gag reflex, abnormal volitional cough, cough after swallow, and voice change after swallow with aspiration, specifically in predicting silent aspiration [47].

In a separate prospective study employing chi-square analysis, researchers investigated factors affecting the 30 days case fatality rate (CFR) among 675 consecutive first time stroke patients. The study revealed substantial disparities in CFR (Case Fatality Rate) among various stroke types. Cerebral infarction showed a CFR of 10%, primary intracerebral hemorrhage had a CFR of 52%, subarachnoid hemorrhage had a CFR of 45%, and patients with an unclear pathological type had a CFR of 74%. Significantly, those with cerebral infarction who were reliant on others for daily functioning prior to experiencing a stroke had a greater likelihood of mortality [14].

Summary: Overall, these researches look into diverse elements of stroke risk, prognosis, and prevention using a variety of statistical techniques, such as Chi-square analysis, logistic regression, and Cox proportional hazards analysis. A compact and concise details of these works are presented in the Table 9 in Appendix B. The findings indicate strong correlations between risk variables such as depression, aortic stiffness, morning home blood pressure, and coronary artery calcification and the incidence of stroke and mortality. Furthermore, the significance of clinical assessments, imaging techniques, and innovative biomarkers in identifying patients at high risk and enhancing risk stratification is emphasized. These observations highlight the significance of thorough evaluation of potential risks and prompt implementation of treatments to prevent and manage strokes. This emphasizes the requirement for holistic methods that consider both physical and psychological health aspects.

7.3 Advancing Stroke Risk Prediction in Atrial Fibrillation: The Role of Computational Physiology and Machine Learning

Recent advances in computational physiology have offered powerful tools to better understand the mechanisms underlying thrombogenesis in atrial fibrillation (AF) and their implications for stroke risk assessment. Traditional risk scores such as CHA₂DS₂-VASc, while widely used, lack the spatial and mechanistic resolution to accurately reflect the dynamic conditions within the left atrium (LA) and its appendage (LAA), where thrombus formation commonly occurs. In this context, several studies have used computational fluid dynamics (CFD) to simulate patient-specific blood flow, uncovering key hemodynamic factors such as low wall shear stress (WSS), high oscillatory shear index (OSI), and relative residence time (RRT), which contribute to blood stasis and clot formation.

Maschi et al. developed a CFD framework using dynamic CT to model LA motion, showing that atrial wall deformation alters flow patterns critical to stroke risk [124]. Paliwal et al. extended this by applying LGE-MRI to map fibrosis, finding lower WSS and higher OSI and RRT in fibrotic regions, with greater disturbances in high fibrotic burden patients, linking fibrosis-driven remodeling to increased stroke susceptibility [147].

Zingaro et al. integrated CFD with functional CMR, combining stroke volume, LA compliance, and flow features to separate stroke patients from controls, with stroke volume normalization improving risk discrimination and supporting fused CFD—imaging approaches for ML models [226]. Gonzalo et al. advanced this through a multiphysics platform integrating electrophysiology, biomechanics, and fluid dynamics in patient-specific LA models, showing fibrosis-induced stiffness and hypocontractility impaired LA and LAA emptying and promoted global flow stagnation, emphasizing the systemic impact of fibrosis on stroke risk [69].

These studies shift stroke risk assessment in AF from population-based heuristics to patient-specific modeling. By linking atrial structure, motion, and flow, computational physiology yields spatially resolved biomarkers for ML-based risk stratification. Markers such as ECAP, OSI, and velocity fields can serve as ML inputs for stroke prediction, enabling probabilistic inference grounded in biophysics. Integrating ML with CFD, particularly in hybrid systems where flow simulations guide learning, offers potential for real-time support in anticoagulation, ablation, and stroke prevention.

In conclusion, the convergence of computational physiology and ML offers a transformative path forward for stroke risk prediction in AF. These models not only explain how fibrosis and altered LA dynamics foster thrombogenesis, but also provide the framework for developing interpretable, mechanistically-informed ML tools that surpass the limitations of current scoring systems. Future efforts should focus on building large-scale datasets that couple imaging, simulation, and clinical outcomes, and on validating predictive performance in prospective cohorts to enable safe and personalized management of stroke risk.

7.4 Machine Learning Approaches

The healthcare sector is evolving with ML, offering advanced analysis of large datasets and accurate predictions. Conventional stroke models such as the Framingham Stroke Risk Profile and CHA₂DS₂-VASc score rely on limited risk factors, assume linearity, and are based on similar populations, restricting their wider applicability. Their predictive efficacy is modest, with C-statistics often between 0.70 and

0.79, and they fail to capture complex multivariate interactions among genetic, clinical, and lifestyle factors. These limitations highlight the need for more flexible and resilient forecasting methods.

ML algorithms—including artificial neural networks, support vector machines, random forests, and decision trees—demonstrate proficiency in identifying complex patterns and nonlinear relationships within extensive, high-dimensional datasets, while progressively enhancing their performance through exposure to fresh data. This skill results in markedly improved prediction of stroke risk in medical diagnostics. Utilising varied patient data, such as medical histories, diagnostic results, and genetic information, ML models enable healthcare providers to make better educated judgements, facilitating early interventions and more tailored treatment plans.

A key distinction in ML approaches lies between supervised and unsupervised learning. Supervised learning uses labelled datasets (patients with and without stroke outcomes) to train models that can predict future cases based on known patterns. This is highly effective in stroke risk prediction, where outcome data are available. Unsupervised learning, by contrast, analyses unlabelled data to uncover hidden structures or clusters (grouping patients by risk profiles without prior outcome labels). Both paradigms are valuable: supervised learning drives predictive accuracy, while unsupervised learning can reveal novel patient subgroups or risk factors not previously considered.

Artificial Neural Networks: A computational model that draws inspiration from biological neural networks is known as an ANN. A neural network is composed of interconnected layers of nodes, known as neurons, which analyze input data using weights (numerical parameters that determine the importance of inputs) and biases (offset values that help adjust model predictions). These neurons learn patterns through backpropagation, an iterative optimization process in which errors are propagated backward through the network to update weights and biases. ANNs are highly adaptable and find use in a wide range of applications such as natural language processing, image recognition, and prediction tasks [21].

A functional ANN model to improve thromboembolic stroke diagnosis was presented in [177], trained on 50 patients using Backpropagation, achieving 89% overall prediction accuracy, with results verified by medical experts, demonstrating precision in early disease identification and treatment planning. Building on this, [39] applied KDP, ANN, and SVM to predict stroke outcomes using data from 297 patients with nine predictors. Feature selection was performed with Cramer's V test, and prediction models applied RBF-SVM and multilayer perceptron ANN. ANN achieved 81.82% and 85.9% accuracy in training and testing with

AUC values of 0.905 and 0.928, slightly outperforming SVM, which had accuracies of 80.38% and 84.62% and AUC values of 0.899 and 0.91. In another study [182], a decision tree for feature selection, PCA for dimensionality reduction, and a backpropagation neural network were applied to the CHS dataset, yielding a predicted accuracy of 97.7% for stroke disease.

Further analysis using PCA and statistical methods [177] identified hypertension, age, heart disease, and average blood sugar as the most important predictors of stroke. Using EHR data from McKinsey & Company available on Kaggle, with 29,072 patients including 548 stroke cases, a downsampling method balanced the dataset. Three ML algorithms were tested, and the perceptron neural network using the four key predictors achieved the highest accuracy of 78% with the lowest miss rate of 19%.

Support Vector Machine: Support vector machine is a supervised learning technique primarily designed for the tasks of classification and regression. SVM operate by identifying a hyperplane that optimally divides distinct classes within a given dataset. This method optimizes the separation between the data points belonging to distinct classes, providing efficient performance, especially in spaces with a large number of dimensions [152].

The study in [23] investigated predicting symptomatic intracranial hemorrhage (SICH) in acute ischemic stroke patients receiving tPA using ML on CT scans and clinical data from 116 patients, including 16 with SICH. An SVM with clinical severity indicators achieved an AUC of 0.744, outperforming traditional prognostic tools and detecting more SICH cases, demonstrating greater automation and effectiveness in prediction. Similarly, a medical data mining study using records from 112 healthy individuals and 80 stroke patients at Turgut Ozal Medical Centre applied PLR, SGB, and SVM with 10-fold cross-validation, finding SVM to perform best with 0.9789 accuracy (95% CI 0.9470–0.9942) and 0.9783 AUC (95% CI 0.9569–0.9997) [6].

Logistic Regression: In the context of machine learning, logistic regression is regarded as a supervised classification approach. The focus transitions from inference to prediction, along with performance. Logistic regression frequently functions as a benchmark model for comparison with more sophisticated methodologies, such as tree boosting or neural networks. Its computational efficiency and interpretability render it a prevalent and reliable aspect of predictive modeling frameworks. Using LR models, [131] examined the effect of stroke location on cognitive and functional outcomes in 428 ischemic stroke patients assessed with MRI. At 3 months, outcomes were measured using mRS and MoCA. Univariate analysis identified stroke site, age, initial NIHSS scores, and stroke volume as predictors of poor outcomes, while multivariate analysis showed stroke location had the

greatest influence on cognitive results (MoCA), enhancing prediction accuracy. However, stroke location did not independently affect functional outcomes (mRS), which were primarily determined by baseline NIHSS scores.

In predicting stroke outcomes, Zihni et al. compared ML techniques, including tree boosting and MLPs, with logistic regression (LR) for forecasting dichotomized 90-day mRS using clinical characteristics, finding similar performance across models with LR (AUCs 0.82, 0.82, 0.79), tree boosting (0.81), and MLP (0.81) [225]. In further research, Lip et al. analyzed medical databases of over 3.4 million individuals with chronic conditions over two years to predict stroke risk, showing that while CHADS₂, CHA₂DS₂-VASc had intermediate accuracy, the multimorbid index had superior discriminant validity, and ML methods such as gradient boosting, ANN, and LR achieved the highest accuracy and greater clinical usefulness [115].

Naive Bayes: Based on Bayes' theorem, a NB model is a probabilistic classifier that assumes the characteristics are conditionally independent from the class label. Although this robust assumption, it excels in classification of text and spam detection tasks for its effectiveness and straightforwardness [162].

Using a variety of physiological variables and ML techniques, the authors [169] trained five distinct models for precise stroke prediction using LR, DT classification, RF classification, KNN, SVM, and NB classification. The research employed the Kaggle Stroke Prediction dataset, which consisted of 5110 rows and 12 columns. In order to enhance user engagement, an HTML page was designed to allow users to input values, while a Flask application was established to establish a connection between the trained models and the web application. Out all the algorithms that were tested, the NB classifier demonstrated the highest performance, achieving an accuracy rate of approximately 82%.

Decision Tree: A DT is a non-parametric computational method employed in supervised learning for the purposes of classification and regression. It partitions the data into subsets according to the values of features, forming a hierarchical structure of decisions and their potential outcomes. The model has high interpretability and adaptability to both numerical and categorical data. Like many machine learning algorithms, DTs can be affected by overfitting if not properly tuned or validated [184], which occurs when the model captures noise in the training data rather than underlying patterns.

The study in [94] aimed to develop a prognostic model for stroke using demographic data and compared DT, NB, and NN classifiers on a resampled dataset of 250 stroke and 500 non-stroke cases. Evaluation metrics included accuracy, FPR, FNR, and AUC. DT achieved the highest accuracy

(0.75) and lowest FPR (0.15), while NN had 0.74 accuracy and the lowest FNR (0.38). NB produced the highest AUC (0.769) but lower accuracy (0.72). The study concluded that DT was most accurate overall, whereas NN offered the best balance of accuracy, FP, and FN rates, making it more suitable for patient safety, and recommended including patients' diagnoses in future models.

Random Forest (RF): A RF is a type of ensemble learning algorithm used for both classification and regression problems. During training, it constructs an ensemble of decision trees and aggregates their results to enhance accuracy and mitigate overfitting. The RF algorithm is designed to handle noisy data and effectively capture intricate relationships between variables by training each tree on a random sample of the data and characteristics [29].

An improvised RF model was proposed in [15] to identify, classify, and forecast strokes from medical data, addressing limitations of prior models. It predicted stroke severity using three hierarchical modules: Module-1 assessed intracerebral and subarachnoid hemorrhagic risk based on ischemic stroke presence, Module-2 on intracerebral hemorrhage, and Module-3 on subarachnoid hemorrhage. With baseline estimation tools and cross-folding for feature selection, the RF classifier achieved 94.23% accuracy, 92.16% sensitivity, 95.07% specificity, and 0.04% error rate. In a related study, [62] applied ML on clinical, biochemical, and neuroimaging parameters of 6022 stroke patients (4922 IS, 1100 ICH) to predict three-month mortality and morbidity, identifying NIHSS scores at admission, 24 h, and 48 h, along with axillary temperature, as key predictors. RF achieved accurate forecasting of long-term outcomes. Similarly, [189] used Kaggle's stroke dataset of 5110 entries, applying SMOTE for balance, and developed models with LR, DT, RF, and Voting classifier. The RF algorithm performed best with attaining an accuracy of about 96 %.

The problem of imbalanced datasets was addressed in [201], which aimed to predict heart attacks among ICU stroke patients. The study introduced the undersampling-clustering-oversampling (UCO) algorithm, combining random undersampling, clustering, and oversampling to balance data. Using 46,520 MIMIC-III records of 2488 stroke patients (82 with heart attacks), ML classifiers were trained, with RF performing best at 70.29% accuracy and 70.05% precision. The findings demonstrated that UCO(120) with RF improves prediction of heart attacks in ICU stroke patients, offering potential for better clinical outcomes.

Stacking: The stacking method is an ensemble learning approach that merges the predictions of several base models to construct a more robust model. In order to enhance prediction precision, a meta learner is employed to combine the outputs of the base learners. In general, the meta learner

employs a distinct algorithm from the base learners in order to generate the final prediction [57].

The authors in [55] presented a comprehensive framework for the long term prediction of stroke occurrence risk was established through the development and evaluation of multiple models utilizing ML techniques. The primary contribution of the work is the successful implementation of a stacking strategy that demonstrates exceptional performance, as confirmed by multiple measures like as AUC, precision, recall, F-measure, and accuracy. The experimental findings have shown that the stacking classification method surpasses previous strategies, achieving an AUC of 98.9%. Additionally, the F-measure, precision, and recall all reach 97.4%, while the accuracy stands at 98%.

Weighted Voting: Weighted voting is an ensemble technique that combines the predictions of many models by assigning weights to each model, which quantify their accuracy or significance. A weighted majority vote determines the final prediction, with more dependable models exerting a greater impact on the result [103].

With consideration for variables including hypertension, body mass index, heart disease, average glucose level, smoking status, previous stroke, and age, the research work in [58] tried to improve early stroke prediction by using several ML techniques. A total of ten classifiers were trained, namely LR, Stochastic Gradient Descent, DT, AdaBoost, Gaussian Classifier, Quadratic Discriminant Analysis, Multilayer Perceptron, KNN, Gradient Boosting, and XGBoost. The outcomes of these classifiers were subsequently combined using a weighted voting method, resulting in a remarkable accuracy rate of 97%. The aggregated voting classifier exhibited superior performance compared to the individual classifiers, showcasing high AUC values and the lowest rates of false positives and false negatives [58].

Ensemble Learning: Ensemble techniques integrate several learning models to enhance overall performance. Ensemble approaches minimize bias, variance, and enhance robustness by combining the predictions of many models such as DT and NN [168]. Major ensemble techniques encompass bagging, boosting, and stacking.

The study in [37] compared different prediction models for predicting stroke risk in a large cohort of Chinese adults, including Cox models, ML techniques, and an ensemble method. A novel ensemble method was devised to predict the risk of stroke over a 9 years period. This method combines gradient boosted trees (GBT) and Cox models, and selectively applies either the GBT or Cox model depending on the individual features. The use of this strategy resulted in the highest level of performance, showcasing greater levels of accuracy (76% for men and 80% for women), specificity (76% for men and 81% for women), and positive predictive

value (26% for men and 24% for women) when compared to any individual model.

Summary: The papers covered in this compilation demonstrate ML growing promise for transforming stroke prediction and outcome assessment. The findings highlight both the potential benefits and the obstacles associated with the use of ML in healthcare. Table 10 in Appendix C displays an overview of various studies and their methodologies, and limitations of ML based models. These requirements encompass the necessity for a wider range and larger quantity of datasets in order to improve the resilience and applicability of the model. Moreover, it is essential to prioritize the interpretability of ML algorithms to facilitate their smooth incorporation into clinical practice. Transparent prospective research with rigorous procedures will play a crucial role in maximizing the potential of ML for developing more precise and easily accessible solutions for managing strokes. This paper provides an overview of various studies that aim to uncover the methodologies, data sources, performance, and limitations of ML based models on stroke prediction.

7.5 Deep Learning Approaches

DL, an application of artificial intelligence, is transforming stroke research by providing robust methods for analyzing and interpreting data. These models are capable of revealing intricate patterns from a variety of data sources, but, they still face problems such as the size of the dataset and the capacity to interpret the model [108]. However, the continuous progress and growing accessibility of healthcare data offer potential for revolutionizing stroke care by improving the diagnosis, treatment, and outcomes for patients.

Deep Neural Network: A DNN is a variant of ANN that include many hidden layers positioned between the input and output layers. The ability of deep architectures to acquire intricate representations of data renders DNNs very effective instruments for various tasks such as speech recognition, image classification, and time series forecasting [170].

The work in [34] examined DL, NB, and SVM for stroke prediction utilizing heart patient data to identify risk variables and predict illness occurrence. The findings aid patient awareness and medical decision making. DNNs showed the highest accuracy in forecasting stroke risk, while NB and SVM performed discretely and linearly. The study compared three models using 899 heart disease records with 76 parameters per record. AF symptoms in cardiac patients are a substantial stroke risk factor which shares predictive factors. The study shows that predictive analytical methods, particularly DL, may help predict stroke. DL, utilized in disease predicting and data analysis, uses patient health information to inform decision making. DL is a viable method of deriving actionable insights from large datasets

with reduced dependence on human experience, which is especially useful considering the many risk factors linked to stroke [34]. Later, another work [83] used DNNs and other ML methods like gradient boosting decision tree (GBDT), LR, and SVM to predict stroke occurrence using large scale electronic medical claims (EMC) data. The dataset includes 798,611 patients, 4726 of whom had strokes during 5 years. Algorithm training uses 2007 characteristics after selection. DNN and GBDT outperform LR and SVM in unweighted average recall (UAR), sensitivities, and specificities, with DNN having the highest prediction accuracy. DNN and GBDT have similar stroke prediction AUCs of 0.915 and 0.918. These results point to the possibility of applying cutting edge ML algorithms to EMC data in order to accurately predict the incidence of stroke [83].

Convolution Neural Network: CNNs are specialised DL models specifically developed to handle grid like input, such as digital photographs. CNNs employ convolutional layers to autonomously identify patterns such as edges, forms, and textures, rendering them highly efficient in tasks related to image and video identification [112].

The primary goal of the study in [138] was to create and confirm a predictive model that can accurately evaluate the final volume of lesions in patients with acute ischemic stroke by utilizing acute MRI. A CNNdeep was trained to forecast ultimate imaging results using acute imaging characteristics. A comparative analysis was conducted to evaluate the performance of CNNdeep in relation to other approaches. These methods include CNNTmax, which is a shallow CNN based on the Tmax biomarker, CNNshallow, which is a shallow CNN based on a mixture of biomarkers, a generalized linear model, and ADCthres, which involves thresholding of the apparent diffusion coefficient biomarker. CNNdeep demonstrated superior performance compared to previous approaches, achieving a greater AUC and enhanced contrast. In addition, CNNdeep showcased its capacity to distinguish outcomes by treatment method, displaying a substantial enhancement in predicting the ultimate size of the infarct. The findings indicate that CNNdeep has the potential to offer valuable automated decision support for tailored treatment planning in situations of acute ischemic stroke [138].

A Dense Convolutional Network (DenseNet) is a CNN architecture in which each layer is interconnected with every other layer in a feed forward mechanism. This high density connection architecture significantly decreases the parameter count, facilitates the reuse of features, and enhances the flow of gradients during training, leading to superior performance in image classification tasks [224].

The DenseNet, a data driven classifier, was presented for the purpose of predicting strokes using 12 lead heart rate data [218]. The dataset consisted of 100 typical 12-lead

ECG traces, which were recorded at a rate of 360 Hz. The dataset was evenly split between normal persons and stroke patients. By carefully adjusting the model, the researchers got outstanding outcomes, with a training accuracy of 99.99% and a prediction accuracy of 85.82%. In addition, the study presented the initial stroke ECG dataset that is accessible to the public, emphasizing the clinical importance of ECG in diagnosing strokes. These findings highlight the potential of using ECG based methods for detecting and diagnosing strokes at an early stage [218].

Long Short Term Memory (LSTM): The LSTM is a recurrent neural network (RNN) specifically developed to exploit long term relationships in sequential data for learning purposes. The system integrates memory cells equipped with gates to regulate the transmission of information, enabling it to selectively retain pertinent data over extended periods of time, rendering it well suited for applications such as time series forecasting and natural language processing [72].

In their work, scientists created a stroke prediction system that combines AI and real time biosignals [221]. The system utilized the RF algorithm for ML and the LSTM approach for DL, incorporating both ML and DL techniques. During routine activities, we gathered real time Electromyography (EMG) signals from the thighs and calves. We then identified significant characteristics to construct a model. The system demonstrated impressive prediction accuracies, with 90.38% accuracy for RF and 98.958% accuracy for LSTM. The findings indicate that the suggested method has the potential to function as a cost effective and real time diagnostic tool for predicting strokes. Additionally, it might also be used to other conditions, such as heart disease.

Gated Recurrent Unit (GRU): The GRU is a streamlined iteration of the LSTM, designed to specifically address the issue of disappearing gradients in RNN. The GRU employs a reduced number of gates and parameters compared to LSTM, therefore enhancing its computational efficiency while effectively capturing dependencies in sequential input. It shows broad application in areas such as time series prediction and language modeling [38]. The system utilizes advanced DL techniques such as LSTM, biLSTM, GRU, and FFNN to effectively handle predictions based on time series data, enabling informed decision making. Bioelectrical signals, including EEG, ECG, and EMG, were obtained using intelligent sensing devices, albeit the precise data is limited. The results demonstrate high accuracy rates, with the GRU model attaining 95.6%, the biLSTM model achieving 91%, the LSTM model reaching 87%, and the FFNN model recording 83%. The results highlight the potential of using time series based methods with advanced computational techniques to improve stroke diagnosis and prognosis.

Recurrent Neural Network: A RNN is a type of neural network specifically developed for processing sequential data. In RNNs, each output is influenced not only by the current input but also by the inputs that came before it. RNNs possess adaptive loops that enable the retention of information, rendering them highly suitable for applications like as language processing and time series analysis. However, they encounter difficulties in handling long term dependencies without adaptations like LSTM or GRU [220].

7.6 Hybrid Models

CNN+LSTM: A novel approach is that uses real time EEG sensor data to allow DL models to be applied on raw EEG data immediately, independent of frequency features is presented in [36]. The researchers applied and contrasted different DL structures, such as LSTM, Bidirectional LSTM, CNN-LSTM, and CNN bidirectional LSTM. The experimental results demonstrated that by employing raw EEG data in conjunction with the CNN bidirectional LSTM model, stroke prediction could be achieved with an accuracy of 94.0%. Furthermore, this approach yielded a low FPR of 6.0% and a FNR of 5.7%. The results highlight the capacity of non invasive techniques that exclusively measure brain waves to accurately forecast and track stroke conditions in real time during everyday tasks [36]. Later, ML methods were employed to analyze real time multi-modal biosignals derived from ECG and photoplethysmography (PPG) measurements [222]. This study presented a system designed to forecast and semantically interpret symptoms associated with stroke prognosis, with a particular focus on the elderly demographic. The stroke illness prediction system incorporates an ensemble architecture that integrates CNN and LSTM models. Data collection was made possible by using a wearable device that had bio-signal sensors. This device allowed for the collection of ECG signals from three electrodes and PPG signals from the index finger. The sampling rate was 1000 Hz per second, and the data was collected during walking activities. The prediction models were assessed for their accuracy. The C4.5 decision tree achieved an accuracy of 91.56%, the RF achieved 97.51%, and the proposed CNN-LSTM model achieved an accuracy of 99.15%. The results demonstrate the effectiveness of the CNN-LSTM model in properly forecasting stroke prognosis using real time multi modal bio-signals [222].

LSTM-RNN: Using EHRs, the study in [35] examined the predictive power of RNNs equipped with LSTM hidden units for stroke prediction. ICD-10 codes and other EHR risk indicators are added to the prediction model. The research employed de-identified EHR files from the Department of Medical Services, Ministry of Public Health of Thailand, covering the years 2014 to 2015. The files included 326,152

records with demographic information, ICD-10 codes, diagnosis dates, clinic kinds, and diagnostic categories. LSTM-RNN performed best for stroke prediction, with accuracy rates of 0.9279, 0.9493, and 0.9998, and F1 scores of 0.9626, 0.9738, and 0.9999. The results imply that LSTM-RNN is more accurate than backpropagation and ordinary RNN algorithms when sample size is greater [35].

RF+DNN: The work carried out in [116] presented a hybrid ML method to forecast cerebral stroke for clinical diagnosis, specifically targeting the difficulties posed by inadequate physiological data and imbalanced class distribution. The methodology consists of two primary steps: first, using RF regression to fill in missing values, and second, deploying an automatic hyperparameter optimization (AutoHPO) based on DNN to forecast strokes on an unbalanced dataset. The medical dataset consists of 43,400 records, which includes 783 instances of stroke. The proposed methodology attains a FNR of 19.1%, which is on average 51.5% lower than traditional methods. Additionally, it exhibits a FPR of 33.1%, an accuracy of 71.6%, and a sensitivity of 67.4% [116].

CNN+NN: By predicting stroke patients' functional outcomes, various DL models were tested to aid intravenous thrombolysis decision making [12]. Diagnostic information and noncontrast computed tomography brain images from intravenous thrombolysis patients were obtained from two tertiary institutions over 7 years. CNN and ANN were used to predict 24 h NIHSS24 and 90 days mRS90 improvement. The best DL model for predicting mRS90 had an accuracy of 0.74 and an F1 score of 0.69, while the best for NIHSS24 had 0.71 and 0.74. These results imply that DL models have potential for functional thrombolysis outcome prediction, indicating the need for more research using bigger datasets and more imaging sequences [12].

Summary: Finally, the study of DL techniques in healthcare shows potential for enhancing the precision of diagnoses and the ability to predict outcomes. Researchers have successfully addressed issues in time series based predictions and image recognition tasks by employing advanced algorithms such as LSTM, biLSTM, GRU, and CNN. The emergence of frameworks such as DenseNet for stroke prediction and CNN bidirectional LSTM for acute ischemic stroke prognosis demonstrates the capacity of DL to facilitate substantial progress in medical research and clinical practice. Table 11 in Appendix D displays the comparative overview of various studies on stroke prediction using DL. The results highlight the significant influence of DL on healthcare decision making processes, providing improved understanding for disease detection, prognosis, and treatment planning. As research in this field progresses, the use of advanced DL techniques shows potential for transforming

healthcare delivery and enhancing patient outcomes on a larger level.

7.7 Emerging Tools for Stroke Risk Assessment and Care Coordination

Recent advancements in AI and DL are significantly advancing healthcare systems' capacity to predict, prevent, and manage chronic and acute health conditions. The SmartCHANGE initiative, a component of the Horizon Europe program, illustrates this transition towards intelligent, proactive healthcare. It addresses a critical concern in public health—the early detection and prevention of non-transmissible chronic illnesses (NCDs) in children and adolescents, including cardiovascular and metabolic conditions. SmartCHANGE utilizes AI-driven decision-support systems informed by extensive longitudinal health statistics. These systems employ federated learning to maintain data privacy among institutions, utilize DL techniques for precise risk prediction, and implement participatory design methodologies to ensure that the resulting tools are easy to use, transparent, and practical for doctors, children, and their families. SmartCHANGE seeks to enhance clinical decision-making while empowering families and health providers to act proactively, finally modifying disease progression and strengthening long-term health outcomes [158].

Concurrently, the InteropEHRate initiative tackles a significant issue in contemporary healthcare: the fragmentation and inaccessibility of health data, especially across European borders. InteropEHRate aims to re-establish individual control over health records via mobile-based Smart EHR (S-EHR) applications. These applications are developed in accordance with HL7 FHIR standards and include GDPR-compliant protocols—including Device-to-Device (D2D), Remote-to-Device (R2D), and Research Data Sharing (RDS)—to guarantee the secure sharing, access, and use of health data for treatment, emergency, or research purposes. InteropEHRate illustrates the synergy between interoperability and patient empowerment in fostering more responsive, personalised, and egalitarian healthcare systems [100].

In addition to extensive public health programs, recent advancements in intelligent healthcare systems (IHS) demonstrate positive outcomes in the application of AI to clinical situations, including stroke prediction. A unique IHS has been created utilising a real-time, web-based application that employs sophisticated DL models—namely, LSTM, GRU, and Bidirectional LSTM (BiLSTM) architectures. Utilising health data from 4,981 individuals, these models assess various personal and clinical characteristics (e.g., age, BMI, glucose levels, heart disease, smoking status) to forecast the probability of a stroke. The BiLSTM model

exhibited outstanding performance, attaining 100% accuracy, precision, recall, and F1-score across all evaluation measures [100].

Collectively, efforts such as SmartCHANGE, InteropEHRate, and the IHS for stroke prediction exemplify a disruptive evolution in healthcare, characterised by the integration of data-driven, AI-enhanced tools with patient-centric methodologies to increase prevention, prediction, and therapy. These initiatives underscore the essential significance of integrating technological advancement with ethical design and user empowerment to develop healthcare systems that are not only more intelligent but also more inclusive, secure, and equipped for future problems. Given that stroke management increasingly depends on early risk evaluation and swift intervention, AI-based solutions may become vital elements of a more proactive and individualised care approach.

7.8 Performance Evaluation of Prediction Models

The performance evaluation matrix holds significant importance in the field of stroke management as it provides a systematic framework for evaluating the efficacy of diagnostic tools, predictive algorithms, remedies, and outcome measures. In the last three decades, there has been substantial progress in evaluating the performance of predictive models in stroke research. Various kinds of metrics are employed to measure model performance, each providing unique insights on diagnostic precision and reliability.

Accuracy measures the ratio of correct predictions to total cases, however it can be deceptive in imbalanced datasets. Sensitivity (recall) measures the model's efficiency in reliably identifying actual stroke instances, whereas specificity rates its effectiveness in correctly detecting non-stroke cases. Precision is the ratio of accurately predicted stroke cases to all positive predictions, providing insight into the model's dependability in a clinical context. The F1 score, denoting the harmonic mean of precision and recall, is especially valuable when it is essential to balance false positives and false negatives. The area under the receiver operating characteristic curve (AUC-ROC) offers a comprehensive assessment of a model's discriminative capability across all classification thresholds [70, 196].

Metrics like as positive predictive value (PPV) and negative predictive value (NPV) are particularly pertinent in assessing the practical utility of models in patient risk evaluations. Enhanced statistical metrics, such as the Matthews correlation coefficient (MCC) and mean squared error (MSE), provide a more equitable and thorough evaluation of model efficacy. Moreover, the concordance index (C-index) is frequently employed in survival analysis to assess the model's efficacy in ranking patients according to risk [98, 196].

These measures collectively provide a sophisticated and multifaceted evaluation of ML models, facilitating their enhancement and possible application in clinical practice.

Statistical Analysis Summary: Table 4 shows that the reviewed studies employed various types of analytical techniques in four major categories. Out of the 228 papers reviewed, 140 are summarized in Table 4 because they directly fall into the listed categories (Traditional statistical methods, Clinical scoring tools, Imaging & psychosocial methods, and Machine learning methods). The remaining papers are discussed in the main text but were not included in the table, as they focus on other aspects (stroke facts, reviews, surveys, treatment, risk factors, and practical implications). Traditional statistical methods, including the Cox Proportional Hazards Model and Multivariate Logistic Regression, have been frequently employed for survival and risk analysis. Clinical scoring tools, such as the NIH Stroke Scale and Modified Rankin Scale, facilitate the assessment of stroke severity and outcomes. Imaging and psychosocial methods, including CT imaging and MRI, offered supplementary predictive features. ML methods were predominantly utilized, with Random Forest, Logistic Regression, and Support Vector Machines being the most common. These methods were assessed using metrics such as accuracy, AUC-ROC, and F1 score. Table 4 illustrates the methodological diversity and the growing significance of data-driven models in stroke research.

8 Discussion

8.1 Key Findings

This research highlights four significant findings in the context of stroke. Comprehending the interaction of lifestyle, genetics, and environmental factors is crucial for reducing stroke risk. Furthermore, significant advancement in stroke prediction has occurred, especially through the utilization of ML models and statistical techniques. These algorithms enhance the precision of stroke incidence predictions, facilitating timely medical interventions. The work underscores the significance of biomarkers in stroke recovery, providing critical insights into the molecular mechanisms that affect healing and rehabilitation. Biomarkers can indicate the extent of harm and the likelihood of recovery, informing individualized therapy approaches [137]. Ultimately, the incorporation of technology ranging from sophisticated imaging methodologies to AI facilitated data analysis has transformed stroke diagnosis and therapy, offering new possibilities for doctors and researchers to improve the treatment of patients. These findings emphasize the necessity of a comprehensive strategy that integrates risk factor analysis,

predictive modeling, biomarker identification, and technological advancement to enhance stroke outcomes.

Risk Factors for Stroke: Several established risk factors elevate the likelihood of stroke occurrence. Conditions like diabetes mellitus contribute to vascular damage, increasing the risk. Psychosocial factors, particularly depression, not only raise stroke risk but also impede recovery by affecting lifestyle and physiological responses [44]. Genetic predispositions, reflected in family history, also play a role [99]. Moreover, cardiovascular related factors such as left atrial enlargement, chronic hypertension, atrial fibrillation, and myocardial infarction have strong associations with stroke risk. Elevated plasma fibrinogen levels further contribute to this by promoting clot formation and inflammation [79, 122].

Advances in Machine Learning for Stroke Prediction: In recent years, ML models have significantly improved stroke prediction and management [37, 62, 114, 115, 189]. DL approaches, including CNNs and LSTM models, show exceptional performance in predicting stroke outcomes based on clinical and imaging data. These models frequently outperform traditional methods, providing enhanced accuracy and more personalized treatment approaches [12, 35, 138, 221, 222]. For instance, CNN based models can predict final lesion volumes in ischemic stroke patients with high accuracy, informing treatment strategies [138].

Biomarkers and Recovery Prediction: Blood biomarkers such as plasma fibrinogen, urinary metabolites, and specific gene expression profiles have emerged as valuable tools for enhancing stroke diagnosis and prediction. These biomarkers can differentiate between stroke subtypes and offer additional insights into patient outcomes [137]. Recovery prediction has also improved with clinical scales like the OPS, which shows superior performance in forecasting functional outcomes in mild to moderate stroke cases compared to traditional scoring systems [207].

Integration of Technology in Stroke Management: The integration of ML models with traditional clinical tools is streamlining stroke risk assessment and resource allocation in healthcare settings. Predictive models not only help estimate long term stroke risks but also assist in decision making processes for personalized patient care. These advancements reflect the growing role of technology in optimizing stroke management and improving patient outcomes. The combination of ML models, advanced biomarkers, and improved recovery prediction tools offers a comprehensive approach to stroke prevention and treatment [137, 167, 186]. These innovations highlight the potential for personalized care and better clinical outcomes for stroke patients.

Summary: The primary findings obtained from the assessment of stroke prediction and management technologies highlight the need of tackling different risk factors

Table 4 Summary of methods, tools, and evaluation metrics across different categories

Category	Method/tools (no. of studies)	Evaluation metrics
Traditional statistical methods	Cox Proportional Hazards Model (7)	Hazard Ratio, Survival Analysis
	Multivariate Logistic Regression (5)	Odds Ratios, p-values, model fit
	Linear Regression (3)	Coefficient significance, R ² value
Clinical scoring tools	Correlation Analysis (2)	Correlation coefficients, p-values
	Stepwise Logistic Regression (2)	Adjusted Odds Ratios, model selection accuracy
	Hazard Ratio / Survival Analysis (3)	Kaplan-Meier curves, HR, CI, time-to-event
	Framingham Risk Score (4)	10-year stroke probability, risk stratification
	NIH Stroke Scale (6)	Severity score, correlation with ADL and outcomes
	Modified Rankin Scale (5)	Disability level (0–6), functional outcome
	Barthel Index (2)	Independence in ADL, functional recovery score
	ASTRAL Score (2)	Stroke severity, mortality prediction
	THRIVE, HIAT, SPAN-100 (2)	Predictive scores for outcome & mortality
	ABCD ² / ABCD ³ / ABCD ³ -I (2)	TIA risk stratification, short-term stroke risk
Imaging & psychosocial methods	Stroke Riskometer (2)	Personalized risk prediction
	Bologna Outcome Algorithm (1)	Recovery outcome assessment
	CHADS ₂ / CHA ₂ DS ₂ -VASc (2)	Stroke risk in atrial fibrillation patients
	CT Imaging / ASPECTS Scoring (4)	ASPECTS score, infarct volume
	MRI with IPH Detection (3)	Lesion characteristics, outcome prediction
	Diffusion/Perfusion Imaging (2)	ADC values, prediction sensitivity
	EEG-Based Prediction (2)	Signal features, classification accuracy
	Retinal Image Analysis (2)	Vessel diameter metrics, ML classification scores
	Carotid Ultrasound / IMT (1)	Intima-media thickness, stroke risk correlation
	EMG-Based Real-Time Prediction (1)	Muscle signal pattern
Machine learning methods	Biomarkers (3)	Protein/fibrinogen levels, statistical correlation with outcomes
	Random Forest (13)	Accuracy, Precision, Recall, F1 score, AUC-ROC, Specificity
	Logistic Regression (10)	
	Support Vector Machine (9)	
	Neural Network (8)	
	Deep Neural Network (5)	
	Long Short-Term Memory (4)	
	Convolutional Neural Network (3)	
	Ensemble / Stacking / Weighted Voting (3)	
	Decision Tree (3)	
	Naïve Bayes (2)	
	AutoML / AutoHPO (2)	
	XGBoost / Gradient Boosting (3)	
	GRU (1)	
	LightGBM (1)	
	Spiking Neural Network (1)	
	CHI / Feature Selection Tools (1)	
	Hybrid Models (3)	

using both conventional and advanced techniques. Although conventional clinical instruments effectively evaluate prevalent risk factors such as hypertension and diabetes, they frequently lack the accuracy required for personalized interventions. Advanced ML techniques have significantly improved stroke prediction by integrating sophisticated datasets, such as biomarkers and imaging data, enabling more precise evaluation of risk and forecasts of recovery [167]. Furthermore, the incorporation of biomarkers into these models has enhanced the capacity for predicting stroke subtypes and patient recoveries. The continuous integration of emerging technologies, such as DL and multimodal data processing, presents encouraging prospects for enhancing the personalized and accurate provision of stroke care. Nevertheless, there are still persistent issues with the generalizability and practical implementation of these advanced methods. Therefore, ongoing study and technical improvement are necessary to fully achieve their promise.

8.2 Comparative Analysis of Approaches in Stroke Prediction and Management

The researchers outline several approaches to stroke prediction and management, focusing on traditional clinical methods and ML based models. By examining these methodologies, it is possible to identify similarities, differences, and inconsistencies, offering insights into their potential strengths and limitations.

8.2.1 The Similarities in Approaches

Focus on Risk Factors: Both traditional methods and ML approaches emphasize the importance of understanding and identifying risk factors for stroke. Risk factors such as diabetes, hypertension, atrial fibrillation, and lifestyle are commonly utilized across all models to predict stroke occurrence. Whether using statistical models or ML algorithms, these factors serve as foundational data for stroke prediction [62, 219].

Emphasis on Personalized Treatment: Both ML models and traditional clinical tools aim to provide personalized care for stroke patients. For example, ML models predict patient specific outcomes, while clinical scales like the OPS are tailored to individual functional recovery, especially in mild to moderate stroke cases [56]. Personalized treatment, thus, remains a key goal across methodologies.

8.2.2 The Differences in Methodologies

Data Usage and Model Complexity: Traditional clinical methods, such as the OPS or the use of biomarkers, rely on predefined clinical parameters or blood markers [56,

137]. These approaches utilize relatively straightforward statistical methods to draw conclusions. In contrast, ML models such as CNNs and LSTMs handle large datasets, including complex imaging data and longitudinal clinical records [221, 222]. ML models process these vast data streams to uncover patterns that might not be evident through traditional approaches.

Accuracy and Outcome Prediction: ML models are demonstrated to be more accurate in stroke outcome prediction compared to traditional methods. DL models can predict final lesion volumes and functional outcomes with high precision, outperforming clinical scales that have been in use for years. CNN based models particularly excel in predicting ischemic stroke outcomes, offering more reliable results than clinical scales or risk score tools [12, 36, 222].

Biomarkers versus Imaging Data: Traditional methods often incorporate biomarkers such as plasma fibrinogen or urinary metabolites to predict stroke risk and subtype [79, 122]. These biomarkers are biological indicators that assist in diagnosis but are somewhat limited in predicting outcomes beyond the initial diagnosis. ML models, on the other hand, leverage imaging data (e.g., MRI scans) alongside clinical data to predict the extent of brain damage, which enhances the overall outcome prediction [137, 197].

Comparative Effectiveness: Machine Learning versus Traditional Methods

Machine learning is increasingly important in stroke care, assisting clinicians make more accurate and timely decisions. To understand the context of these advancements, we present a comparative summary (Table 5) that evaluates its performance against traditional methods in areas such as diagnosis, outcome prediction, treatment selection, and patient risk assessment.

Note: The reported improvements are derived from recent research and observational evaluations. Randomized controlled trials remain necessary to confirm these improvements in actual clinical outcomes.

8.2.3 Inconsistencies and Contradictions in Findings

Accuracy and Generalizability of ML Models: While ML models show superior accuracy, some concerns about their generalizability are evident. Research suggests that while DL models work well in controlled settings with ample data, they may not perform as reliably in real world scenarios where patient data are more variable [36, 222]. Additionally, there is inconsistency in how well these models can be applied across diverse populations due to variability in stroke presentations globally. This contrasts with traditional methods, which, while less precise, have broader applicability due to their simplicity and widespread validation.

Table 5 Comparative evaluation of traditional clinical methods vs machine learning in stroke care

Domain	Traditional Clinical Methods	Machine Learning (ML) Approaches	Improvement Brought by ML
Diagnosis (Imaging)	Radiologist interpretation of CT/MRI; subjective and time-consuming	Automated image analysis using CNNs and DL models	Faster detection, higher sensitivity (e.g., early infarct detection)
Stroke Subtype Classification	TOAST-based classification by clinician judgment	ML classifiers using imaging + clinical data	More precise classification, improved inter-rater reliability
Outcome Prediction	Risk scores (e.g., I-Score), regression models	ML models integrating multidimensional data (labs, EHR, demographics, imaging)	Higher predictive accuracy for 30-day and 1-year mortality
Treatment Decision Support	Time-based eligibility for tPA/thrombectomy	ML-guided risk-benefit profiling for reperfusion therapy	Enhanced patient selection, reduced over/under-treatment
Risk Stratification	Manual scoring (e.g., hypertension, AF, diabetes)	Ensemble models (e.g., random forests) evaluating complex risk interactions	Better identification of high-risk patients for recurrence or complications
Workflow Efficiency	Clinician-dependent, time-intensive processes	Real-time alerts, triage tools integrated into EHR systems	Reduced time to diagnosis and treatment; improved resource allocation

Recovery Prediction: One potential contradiction lies in recovery prediction. While ML models demonstrate accuracy in predicting lesion volume and stroke outcomes, our research notes that traditional scales like the OPS are still highly effective, particularly in mild to moderate cases [56]. This raises the question of whether ML models offer

significant improvement over well established scales for certain patient categories, or if traditional methods suffice for these cases.

8.2.4 Potential Reasons for Inconsistencies

Data Quality and Availability: One of the primary reasons for inconsistencies in findings is likely the availability and quality of data. ML models require large, high quality datasets to function optimally. In contrast, traditional methods, which are less data intensive, can be applied more universally. When ML models are trained on limited or unrepresentative data, their performance may decline, leading to discrepancies between clinical settings and controlled trials.

Complexity Of stroke Presentations: The inherent variability in stroke presentations, including differences in age, comorbidity, and stroke subtypes, complicates prediction models. Traditional methods may not capture the full complexity of these presentations, but they are simple enough to apply broadly. In contrast, ML models can capture more complexity but may struggle with outliers or cases that do not fit typical patterns, causing inconsistencies in performance.

Bias in Machine Learning Algorithms: Bias in ML algorithms may also contribute to contradictions. Models trained on datasets that are not diverse may inadvertently favor certain patient demographics or stroke presentations. This is a concern when deploying ML based solutions in real world healthcare settings, where patient populations are heterogeneous [32, 85].

Summary: The comparison of traditional and ML based approaches to stroke prediction and management in Table 6 reveals both similarities in their objectives and significant differences in methodology and performance. (Note that the accuracy values presented are representative data obtained from the literature, designed to demonstrate typical performance trends across various methodologies. The measurements are neither absolute nor generally applicable, as performance fluctuates based on dataset attributes, study design, and outcome definitions. The table aims to offer a comparison approach that emphasizes relative strengths, limitations, and prevalent difficulties instead of exact aggregated information.) While ML models offer greater precision and the potential for personalized treatment, traditional methods remain valuable due to their simplicity, accessibility, and broad applicability. Inconsistencies in the findings, especially regarding the generalizability and recovery prediction accuracy of ML models, are likely tied to data quality and inherent biases in ML algorithms. As the integration of ML into clinical practice advances, addressing these inconsistencies through more robust and diverse datasets will be essential for optimizing stroke care. This

Table 6 Comparison and contrast of several methodologies explored in this study

Aspects	Machine learning models	Traditional Models	ML + Biomarkers	Traditional + Biomarkers	Long term outcome prediction	Inconsistencies	Reasons for inconsistencies
Accuracy	High (95%)	Moderate (80%)	High (92%)	High (85%)	Low (70%)	Traditional methods underperform in complex cases	Traditional methods rely heavily on clinical scales without advanced data analysis
Generalizability	Low	High	Medium	High	Low	ML models struggle to generalize to new populations	ML models trained on homogeneous datasets and lack diversity
Data input	Large datasets needed	Small datasets based on clinical scales	Biomarkers + ML	Biomarkers + Clinical Data	Mixed input types	ML + Biomarkers needs more extensive data	Biomarker integration is not standardized across models
Interpretability	Low	High	Medium	High	Medium	ML models are often “black boxes” and hard to interpret	Complexity of ML algorithms, lack of explainability in AI models
Cost effectiveness	Medium	High	Low	High	Low	ML and biomarker approaches require more resources	Computational and data related costs make ML approaches expensive
Training requirements	High	Low	High	Low	Medium	ML approaches require technical expertise	ML models require specialized knowledge to be used effectively
Recovery prediction	Limited	Strong for short term recovery	Strong	Strong	Focus on long term outcomes	ML models don't focus much on long term recovery	ML models prioritize immediate outcomes over long term recovery predictions
Dataset size requirement	High	Low	High	Low	Medium	ML models require vast datasets for training	ML models need large scale data for accuracy and generalization
Bias risk	High (due to data homogeneity)	Low	Moderate	Low	Moderate	ML models may reflect training dataset biases	Inadequate representation of different demographic groups in datasets
Contradictions in methodology	ML models outperform traditional scales but struggle with diverse populations	Traditional scales are easier to use but less accurate	Combination of biomarkers improves accuracy but raises cost	Traditional + Biomarkers approach is consistent and practical	Long term prediction models are less accurate	Traditional approaches are favored for short term, but ML is better for complex predictions	Traditional methods are tested on broader populations, while ML models are still in experimental stages with narrow testing scope

comparison can guide future research to address the identified gaps and inconsistencies.

8.3 Strengths and Limitations of Current Approaches

Our study outlines a range of approaches to stroke prediction and management, focusing on both traditional clinical methods and ML based models. Each of these approaches offers unique strengths and faces inherent limitations, which ultimately influence the overall conclusions drawn regarding stroke care.

8.3.1 Strengths and Weaknesses of Statistical Methods

Statistical methods have long been central to stroke prediction and management due to their simplicity, transparency, and ease of interpretation. These approaches require relatively small datasets, making them practical and cost-effective in both high- and low-resource settings. Furthermore, they are widely validated in clinical practice and provide clinicians with straightforward tools for estimating risk, particularly through established scoring systems. Statistical approaches are also useful for identifying associations between risk factors and stroke outcomes, offering valuable insights into population-level trends.

Despite these advantages, statistical methods face several limitations. Their predictive accuracy is often reduced in heterogeneous or complex patient populations, as they are

unable to adequately model non-linear relationships among variables. Moreover, these methods lack the capacity to effectively integrate multidimensional data such as imaging, genetic, or biomarker information, which increasingly play a role in stroke risk assessment. Consequently, predictions based on statistical methods tend to be generalized rather than individualized, limiting their utility for precision medicine. Additionally, their performance in predicting long-term recovery outcomes is less robust compared to more advanced techniques.

8.3.2 Strengths and Weaknesses of Machine Learning Methods

Machine learning (ML) and deep learning approaches represent a significant advancement in stroke prediction, offering the ability to process large, complex datasets with greater predictive accuracy than traditional methods. These models can capture non-linear and hidden relationships between variables and are capable of integrating diverse data types, including clinical records, biomarkers, imaging, and genetic information. By leveraging these multidimensional inputs, ML models enable more personalized and precise risk assessments, aligning closely with the goals of precision medicine. Furthermore, such models can continuously improve as additional real-world or real-time data becomes available, enhancing their adaptability and long-term predictive utility.

Nevertheless, ML approaches are not without limitations. Their reliance on large, high-quality datasets poses a barrier, particularly in settings where such data are unavailable or inconsistent. Generalizability also remains a challenge, as models trained on homogeneous datasets often perform poorly when applied to diverse populations. Another key limitation is interpretability: many ML algorithms operate as “black-box” systems, making their decision-making processes opaque to clinicians and reducing trust in their outputs. Additionally, these methods demand substantial computational resources and specialized expertise, which restricts their scalability and practical implementation in resource-limited environments.

8.4 Implications of the Findings on Stroke Prediction and Management

The findings of this study offer important contributions to the understanding of stroke prediction and management, revealing both the advancements and limitations of current approaches. These insights shape not only the scientific discourse but also provide practical implications for future research, policy making, and clinical practice.

8.4.1 Contribution to Understanding Stroke Prediction and Management

Integration of Machine Learning in Healthcare: One of the most notable contributions of the findings is the clear potential for ML models, especially DL approaches, to revolutionize stroke prediction and management. By demonstrating that models such as CNNs and LSTMs outperform traditional clinical scales in predicting outcomes and processing complex data, the findings establish ML as a critical component in advancing healthcare technologies. This contribution shifts the field towards data driven, personalized healthcare, underscoring the role of technology in improving patient outcomes [12, 221, 222].

Personalization and Precision Medicine: The emphasis on personalized treatment is a significant insight. ML models that use patient specific data, including imaging and biomarkers, contribute to a more precise form of medicine tailored to individual characteristics. This marks a shift away from generalized treatment protocols towards more nuanced care strategies that consider each patient’s unique risk factors and health profile [137, 197].

Importance of Biomarkers: The findings reiterate the value of biomarkers, such as plasma fibrinogen and urinary metabolites, in stroke prediction. The integration of these biological markers into both traditional and ML models enhances diagnostic accuracy, particularly in differentiating between stroke subtypes. This underscores the role of biomarkers in early detection, risk assessment, and prognosis, which is crucial for timely and effective intervention.

8.4.2 New Insights and Perspectives

The Role of Data in Stroke Prediction: A key new insight from this study is the importance of data quality and diversity in ML model performance. While ML models are shown to be highly accurate in controlled settings, the findings highlight the limitations in generalizability when these models are applied to diverse real world populations. This raises awareness about the need for more inclusive, representative datasets that can address healthcare disparities and avoid biases in prediction models.

Balancing Traditional and Technological Approaches: Another emerging perspective is the recognition that both ML models and traditional clinical tools have unique strengths, suggesting a need for balance rather than outright replacement. This research shows that traditional tools like the OPS remain highly effective in specific cases, such as predicting recovery in mild to moderate stroke patients [106]. This calls for a more integrative approach, where ML models complement rather than replace established methods.

Challenges of Implementing Advanced Models in Clinical Practice: The findings bring to light the challenges of adopting complex ML models in clinical settings. Issues like the need for specialized training, model interpretability, and reliance on high quality data may hinder their widespread use. This insight raises important questions about the practicality of ML models in everyday healthcare and suggests that future research must focus on making these technologies more accessible to healthcare providers.

8.4.3 Practical Implications for Future Research, Policy, and Practice

Future Research on Data Diversity and Bias: The limitations of ML models regarding generalizability suggest that future research should prioritize building more diverse datasets to improve model robustness. Research efforts must focus on collecting data from different populations, including those underrepresented in current datasets, to reduce bias and ensure that ML models are equitable and effective for all patient groups. Moreover, developing explainable AI systems that provide clinicians with understandable outputs could bridge the gap between technological advances and practical application.

Policy Implications for Technology Integration in Healthcare: From a policy perspective, the findings imply a need for new frameworks that guide the integration of ML technologies in healthcare. Policymakers should consider setting standards for the ethical use of AI in clinical settings, ensuring that data privacy, patient consent, and algorithmic fairness are prioritized. Additionally, funding should be directed towards interdisciplinary collaborations between data scientists and healthcare providers to ensure the safe and effective use of ML in stroke management.

Clinical Practice and Workforce Training: The practical implementation of ML models in clinical practice will require investment in training for healthcare professionals. The complexity of these models may present a barrier to their adoption, and healthcare workers will need to be trained in data interpretation, the use of AI tools, and understanding the limitations of these technologies. This calls for updating medical education curricula to include data literacy and AI competencies, ensuring that clinicians are prepared to integrate these tools into patient care.

Resource Allocation and Cost Effectiveness: Integrating ML in limited resources healthcare environments requires initial expenditures, encompassing IT enhancements, data digitisation, secure storage solutions, and people training, particularly in contexts characterised by paper-based or inadequately maintained systems [157, 174]. However, ML provides enduring advantages by facilitating earlier diagnoses, enhancing risk prediction, and optimising

resource allocation, especially in stroke management, where predictive models can decrease hospitalisations and treatment expenses [9, 59]. The cost can be mitigated using cloud-based platforms, open-source tools such as TensorFlow and Scikit-learn, academic collaborations, and by customising pre-trained models to local datasets instead of developing new ones.

Broader Applications of Our Findings: Methods of ML created for stroke prediction have shown significant relevance in both healthcare and non-healthcare settings. In the medical field, these models provide early diagnosis, prognosis, and tailored therapy. For example, whereas CNNs coupled with NLP technologies project Alzheimer's Disease development depending on imaging and speech, decision-tree models estimate mobility assistance needs in Parkinson's Disease. Trained on large-scale EHRs, gradient-boosted models—e.g., XGBoost—forecast diabetes consequences including retinopathy, hence promoting preventative therapy. DL models combining genetic and imaging data can forecast treatment response in oncology, therefore allowing tailored cancer therapy. Heart failure uses logistic regression to project 30-day readmission risk, hence directing home care plans and follow-up. Apart from medicine, other sectors use comparable ML techniques. Models such as XGBoost in finance forecast loan defaults using applicant financial data, therefore allowing automated credit judgments. ARIMA-LSTM hybrids in the energy industry predict power demand, hence enhancing grid control. In disaster response, CNNs help with resource allocation and emergency planning by predicting natural hazard risks from satellite and climatic data.

Summary: The findings from this study significantly enhance the understanding of stroke prediction and management by highlighting the strengths of ML models in improving predictive accuracy and personalizing care. However, they also reveal important limitations, particularly related to data diversity, interpretability, and real world implementation challenges. These insights provide a roadmap for future research focused on addressing these limitations and ensuring that the benefits of technological advancements are equitably distributed across different healthcare settings. Policymakers and practitioners will need to collaborate closely to integrate these findings into future strategies for stroke care, balancing innovation with practicality to achieve the best patient outcomes.

8.5 Strength and Limitation of the Proposed Work

This review examines approaches for stroke prediction and prevention, including a diverse array of evidence from clinical tools and novel machine learning techniques. In any synthesis of a swiftly advancing topic, it is crucial to

evaluate both the contributions and limitations of the effort. Identifying the study's strengths underscores its significance in enhancing knowledge and informing practice, but recognizing its limits offers a pragmatic view on the relevance of its findings and indicates priorities for subsequent research. This section highlights the strengths and limits comprehensively.

8.5.1 Strength

This review addresses a critical global health issue by synthesizing existing research on stroke prediction and prevention, analyzing traditional clinical tools with contemporary machine learning methods. A significant strength is its comprehensive and inclusive approach, incorporating studies from imaging, bioelectrical signals, biomarkers, and tabular health data. This offers a comprehensive view of the contributions of various methodologies to stroke risk assessment and management. A structured PICO-based search strategy enhances transparency and rigor in the selection process. The screening of over 3000 studies, reduced to 228 high-quality articles, highlights the comprehensiveness and reliability of the review.

The study provides a balanced and critical evaluation, which is equally important. This research emphasizes the ongoing relevance of traditional tools, particularly in resource-constrained environments, while advocating for hybrid approaches that combine both machine learning and conventional methods to enhance clinical feasibility. The discussion recognizes challenges including generalizability, interpretability, and ethical concerns, thereby ensuring that the findings are relevant to real-world contexts. The review's forward-looking orientation, featuring recommendations for explainable AI, multimodal data fusion, and fairness-aware algorithms, illustrates its practical significance by connecting scientific advancements to prospective clinical applications and global healthcare requirements.

8.5.2 Limitations

This study does not adhere strictly to PRISMA criteria. Although this may restrict methodological rigor and reproducibility relative to a fully standardized systematic review, the decision was deliberate: the flexibility permitted the authors to incorporate a broader spectrum of study types (including narrative reviews, exploratory analyses, and computational modeling papers) that would otherwise be omitted. The review offers a more thorough overview of the field, however with limited requirements

An additional limitation is the absence of comprehensive quality assessment for each included study. Due to the diverse approaches, including clinical cohort studies

and deep learning model assessments, direct comparison of methodological quality was not consistently achievable. The review highlights patterns and trends among research, offering valuable strategic insights despite the constraints of detailed critical assessment.

The generalizability of findings is a concern as several reviewed machine learning experiments depend on single-institution or demographically limited datasets. Nonetheless, emphasizing this constraint fulfills a crucial role: it highlights the pressing necessity for more diverse, multi-center data to render these models clinically dependable. Challenges concerning the interpretability and ethical implementation of AI systems are recognized not as obstacles to advancement, but as critical domains for future study and clinical application.

Overall, although the synthesis highlights positive outcomes, it assigns less significance to research with poor or equivocal results. This highlights the exploratory character of the review, which seeks to identify emergent opportunities rather than do a rigorous assessment of evidence hierarchy. Recognizing both achievements and obstacles offers a balanced viewpoint and mitigates the exaggeration of machine learning's potential.

8.6 Synthesis of Insights

This review emphasizes a crucial period in stroke prediction and management, wherein traditional clinical tools remain foundational, while ML techniques are progressively revealing their capacity to transform practice. DL algorithms, specifically, have the capability to identify intricate data patterns and facilitate individualized treatment approaches. Nonetheless, the potential of machine learning is mitigated by ongoing uncertainties regarding generalizability, interpretability, and ethical application. The following are several overarching insights:

The Synergistic Value of Machine Learning and Traditional Approaches. Although machine learning can improve predictions, conventional methods are essential due to their familiarity, accessibility, and clinical foundation. Future advancements are expected to arise from hybrid frameworks that integrate the advantages of both.

Data Inclusion and Diversity are Essential Enablers. Existing machine learning models are frequently developed on restricted datasets, which constrains their applicability across varied patient groups. Enhancing databases to encompass demographic, geographic, and clinical diversity will be crucial for developing equitable instruments.

The Primary Impact of Explainability. In the absence of interpretability, even models with excellent accuracy encounter opposition in clinical practice. Creating explainable AI that delivers transparent, actionable information for

clinicians is not merely a technical hurdle but an essential condition for adoption.

Ethical and Regulatory Protections. The incorporation of machine learning in healthcare has substantial concerns of privacy, bias, and equity. Techniques like federated learning and differential privacy, in conjunction with transparent consent and auditing mechanisms, will be essential to reconcile innovation with accountability.

Themes can have direct practical effects. Healthcare personnel must get training in AI literacy to guarantee the responsible and successful utilization of tools. Policymakers will influence the rate and extent of machine learning adoption through the establishment of explicit ethical and regulatory frameworks. Researchers must not only advance the precision of their work but also guarantee the scalability, cost-effectiveness, and inclusivity of models in practical applications.

In conclusion, the synthesis indicates a future in which stroke prediction progresses not solely through technology, but through a collaborative effort encompassing research, practice, and policy. By addressing issues of diversity, openness, and ethical governance, ML can progress from a promising breakthrough to a reliable and revolutionary element of stroke care.

8.7 Identifying Gaps and Future Directions in Stroke Prediction and Management

This research provides an extensive overview of traditional and ML based approaches in stroke prediction and management. However, there are clear gaps that remain in the research and application of these methods, limiting their potential impact on clinical practice. Addressing these gaps will help advance the field and optimize stroke care.

8.7.1 Key Gaps in the Current Research

Limited Generalizability of ML Models: One of the primary gaps identified in our study is the limited generalizability of ML models. These models, while highly accurate in controlled settings, may not perform as effectively across diverse real world populations. This raises concerns about how applicable these models are outside of the specific datasets on which they were trained. Ensuring that ML models can be used in various healthcare contexts and with heterogeneous patient data remains a challenge.

Lack of Diverse and Representative Data: Another gap is the lack of diverse, representative datasets used to train ML models. Many of the current models are based on datasets that may not include patients from different geographic, ethnic, or socioeconomic backgrounds. This could lead to

biased predictions and reduce the accuracy and fairness of these models when applied to broader populations.

Limited Exploration of Biomarkers in ML Models: While biomarkers are highlighted as valuable in traditional stroke prediction methods, their integration into ML models appears to be underexplored. Biomarkers such as plasma fibrinogen and urinary metabolites are important for early diagnosis and stroke subtype differentiation, but their potential to improve the performance of ML models has not been fully realized.

Challenges in Implementation and Interpretability of ML Models: There are challenges related to the interpretability and practical implementation of ML models in clinical settings. Clinicians need clear, actionable insights from ML algorithms, but the complexity of these models makes it difficult to translate their outputs into straightforward clinical decisions. This gap highlights the need for developing more explainable AI models.

Inconsistent Recovery Prediction Accuracy: The findings show inconsistencies in the accuracy of recovery predictions between traditional methods and ML models. Traditional tools like the OPS perform well in certain contexts, such as predicting mild to moderate stroke recovery, but ML models may offer little additional value in these cases. This suggests a gap in understanding which patients benefit most from advanced ML tools [106].

8.7.2 Future Research Directions and Proposes Methodologies to Address these Gaps

Addressing these gaps will require targeted research efforts across several key areas. By exploring new questions and methodologies, future studies can help refine and optimize stroke prediction and management strategies. While the study demonstrates high prediction accuracy using various ML models, particularly neural networks, it does not yet explore how such models can be effectively integrated into real-world clinical workflows. This remains a critical next step for translating research into practice. Future work will involve collaborating with clinical practitioners to understand operational constraints and identify integration points, such as embedding the model into electronic health record (EHR) systems for real-time decision support. Moreover, practical considerations such as model interpretability, clinical validation, regulatory compliance, and ease of deployment must be systematically addressed to ensure clinical adoption. These efforts will help bridge the gap between algorithmic performance and clinical utility, enabling the development of stroke prediction tools that are not only accurate but also usable in day-to-day healthcare settings.

Summary: Table 7 shows the gaps identified in this study related to data diversity, ML model generalizability,

Table 7 The key gaps identified in this study, suggested areas where future research is needed, and propose methodologies to address these gaps

Gap identified	Area for future research	Proposed research methodologies
Limited generalizability of ML models.	Improving model generalizability across diverse populations.	Multi site studies using federated learning to train ML models across various populations, preserving patient privacy while ensuring diversity in data.
Lack of diverse and representative data (bias in ML algorithms).	Addressing bias in ML models for equitable care.	Develop fairness aware algorithms and use adversarial ML techniques to detect and mitigate bias. Incorporate fairness constraints based on demographic factors.
Integration of biomarkers and imaging data.	Combining biomarkers and imaging for stroke prediction.	Use multimodal DL models to integrate biomarkers and imaging data (e.g., MRI, CT scans) with clinical data for better predictions.
Lack of interpretable ML models	Enhancing interpretability of ML models for clinical use	Develop explainable AI techniques such as SHAP and LIME to ensure ML predictions are understandable and actionable for clinicians.
Limited focus on long term outcome prediction	Predicting long term stroke recovery outcomes	Use longitudinal cohort studies and RNNs to capture temporal data and predict long term functional recovery trajectories.
Cost effectiveness in resource limited settings	Assessing cost effectiveness of ML models in different settings	Conduct cost benefit analyses and implementation studies to assess the practicality of ML models in low resource environments. Explore the development of lightweight ML models.
Insufficient integration of clinical and computational data	Bridging clinical and computational approaches in stroke management	Use hybrid approaches combining clinical expertise with ML based predictions. Incorporate feedback loops between human clinicians and algorithms for improved accuracy.
Gaps in predictive power for atypical cases	Enhancing predictive power in atypical stroke cases	Conduct studies on rare or atypical stroke cases using transfer learning techniques to adapt ML models to underrepresented conditions.

biomarker integration, explainability, and recovery prediction accuracy highlight critical areas where future research is needed. By addressing these gaps, researchers can refine ML models for stroke prediction and management, ensuring that they are more equitable, accurate, and useful in clinical practice. Future studies should focus on creating diverse datasets, integrating biomarkers into models, and improving the explainability and real world feasibility of ML tools. These efforts will advance the field and contribute to more personalized, data driven approaches to stroke care.

9 Conclusion

This review provides significant insights into the strengths, limitations, and future directions of stroke prediction and management techniques, highlighting the vital significance of risk factors including hypertension, diabetes, atrial fibrillation, and lifestyle choices such as smoking and physical inactivity. Risk scores that are widely applicable and straightforward are examples of traditional clinical tools that have proved useful in evaluating these issues. However, they frequently lack the accuracy necessary to consider the complexities of specific patient profiles, especially in integrating several risk factors or addressing heterogeneous populations. These constraints highlight the necessity for more sophisticated approaches in stroke management.

This study significantly contributes by rigorously comparing traditional and machine learning methodologies for stroke prediction, diagnosis, prognosis, and treatment. While traditional tools are essential in low-resource settings for preliminary assessments, machine learning techniques have distinct benefits in handling high-dimensional, multi-modal data. The review highlights the importance of hybrid approaches that utilize the advantages of both paradigms by highlighting the present predicament. This comparative paradigm distinguishes our work from previous studies and offers a guide for future research and practical implementation of stroke prediction and treatment methods.

This research rigorously evaluated the use of ML and DL methodologies in predicting stroke risk, highlighting their capacity to outperform traditional models that frequently depend on linear assumptions and exhibit limited generalisability. ML and DL models possess the potential to analyse extensive and intricate datasets that surpass conventional clinical characteristics, incorporating imaging data, genetic profiles, and biomarkers. These models provide more precise and individualised risk evaluations by recognising complex patterns and correlations sometimes overlooked by conventional methods. Although their practical application is restricted by various limitations, such as restricted generalisability across heterogeneous populations, difficulties in model interpretability that impede clinical integration, and the necessity for extensive, high-quality datasets—resources that may not be universally accessible, especially in resource-limited environments. These limitations underscore the necessity for continuous research focused on augmenting model transparency, guaranteeing equal dataset representation, and strengthening external validation. The results of this study are pertinent to various stakeholders: healthcare professionals may implement ML tools for the early detection of high-risk patients; AI researchers can rectify existing methodological deficiencies to enhance model reliability; policymakers and healthcare administrators can

leverage AI-generated insights to optimise preventive strategies; and patients could experience more personalised care and enhanced outcomes. To facilitate extensive distribution and adoption, these findings will be communicated via scientific conferences, clinical webinars, and professional networks. Ultimately, enhancing the clinical application of ML in stroke care necessitates an integrated endeavour to connect technological advancement with practical execution.

This review emphasizes several critical areas that require further investigation for future researchers. One critical area is expanding the generalizability of predictive algorithms by incorporating a wider range of datasets that more accurately represent a variety of populations, including minority groups. Furthermore, it will be critical to improve the interpretability of these models in order to gain the trust of medical practitioners and facilitate their integration into the clinical environment. The integration of new data sources, genetic information, including biomarkers, and advanced imaging techniques is another prospective possibility that could further improve the accuracy and precision of stroke prediction models.

Addressing these challenges would improve stroke prediction accuracy, provide more efficient preventative efforts, timely interventions, and individualized treatment approaches that could significantly enhance patient outcomes. Future research, using the insights and discoveries outlined in this review, might accelerate the development of innovative stroke treatment tools, thereby improving stroke treatment efficacy, data driven nature, and patient

centeredness. This review research provides a critical foundation for individuals seeking to advance stroke research, aiding in the resolution of persistent challenges in stroke prediction and prevention, and encouraging innovations that will save lives and mitigate the long term effects of stroke.

Appendix A: Glossary of Technical Terms

See Table 8.

Table 8 Definitions of various terms related to medical and machine learning

Term	Definition
ADCthres	Apparent Diffusion Coefficient threshold, a value used in diffusion-weighted imaging (DWI) MRI to differentiate between healthy and damaged brain tissue
Algorithm	A well-defined sequence of instructions or rules used to perform a computation or solve a problem
AutoHPO	Automated Hyperparameter Optimization, the automated process of finding the optimal set of hyperparameters for a machine learning model
Bayes, Bayesian	Relating to Bayes' theorem, a fundamental principle in probability theory that describes how to update the probability of a hypothesis based on new evidence
CI	Confidence Interval, a range of values estimating the true population parameter with a certain level of confidence (e.g., 95%)
CNNTmax	Convolutional Neural Network Tmax, the application of CNNs to analyze Tmax (time to peak) perfusion maps in stroke imaging
CT	Computed Tomography, a medical imaging technique using X-rays and computer processing to create detailed cross-sectional images of the body
Echoplanar	A type of MRI sequence that allows for rapid image acquisition
EMG	Electromyography, a diagnostic test that assesses the health of muscles and the nerves that control them
Ensemble	In machine learning, an ensemble method combines predictions from multiple models to improve overall accuracy
IDI	Integrated Discrimination Improvement, a metric used to assess the improvement in risk prediction when adding a new marker to a model
iScore	Integrated Score, likely refers to a composite risk score combining multiple clinical factors. The precise definition requires more context
LIME	Local Interpretable Model-agnostic Explanations, a technique explaining individual predictions of any machine learning model
MSE	Mean Squared Error, a measure of the average squared difference between predicted and actual values in regression models
Ndeep	Network deep, likely refers to the depth or number of layers in a neural network architecture
PFI	Physical Function Index, a measure of an individual's ability to perform activities of daily living
PPG	Photoplethysmography, an optical technique to detect blood volume changes in the microvasculature
PWI	Perfusion-Weighted Imaging, an MRI technique that assesses blood flow in the brain tissue
RBF	Radial Basis Function, a type of function used in machine learning, particularly in radial basis function networks
Riskometer	A tool or algorithm used to assess an individual's risk of developing a particular disease or condition
SHAP	SHapley Additive exPlanations, a method for explaining the output of machine learning models
Stochas	Stochastic, involving randomness or probability. In optimization, stochastic methods involve random sampling to find solutions
Tmax	Time to maximum, often used in perfusion imaging to describe the time it takes for contrast to reach its peak in a specific brain region
Voxel	A three-dimensional pixel, representing a volume element in a three-dimensional image

Appendix B: Summary of Stroke Risk Studies

See Table 9.

Table 9 An overview of various studies aimed at uncovering risk factors for stroke, their methodologies, key findings, and limitations of statistical analysis-based models

Studies	Objectives	Dataset, study types and methodology	Key findings	Limitations
[33]	To uncover ischemic stroke risk factors and biomarkers, assess their predictive ability, and improve risk stratification algorithms for preventative interventions	Data source (D): The ARIC Study Study type (S): Prospective cohort study Methodology (M): Cox regression model	predictive improvement with non-traditional factors	The study's conclusions may not apply to other demographic and socioeconomic groups due to its cohort from four US cities
[49]	To investigate the relationship between stroke volume/pulse pressure ratio and cardiovascular risk	D: New York hospital–cornell medical center S: Prospective cohort study M: Multiple logistic regression models and cox proportional hazards analysis	The stroke volume/pulse pressure ratio is a potentially valuable marker for cardiovascular risk assessment	The use of peripheral instead of central precocious puberty
[87]	To assess baseline levels of depression as a risk factor for stroke among white and black men and women	D: NHANES and NHEFS S: Prospective cohort study M: Cox proportional hazards models	Depression is predictive of stroke across all classes	Followup loss (5.2%) and the absence of baseline data (3.9%)
[141]	To examine the relationship between 24 h, daytime, and nighttime ambulatory blood pressures and first symptomatic stroke	D: The Ohasama study S: Prospective cohort study M: Cox proportional hazards regression model	Ambulatory blood pressure values were more predictive power than screening blood pressure, and daytime blood pressure was better than nighttime blood pressure	The study's generalizability may be limited due to the inclusion of Ohasama, Japan residents over 40 without symptomatic stroke, potentially excluding more diverse populations
[107]	To determine the correlation between aortic stiffness and mortality due to stroke in patients with hypertension	D: The outpatient hypertension clinic of hospital Broussais S: Predict future outcomes (longitudinal cohort study) M: Cox proportional hazard regression model	This study presents the initial argument, in a longitudinal investigation, that aortic stiffness is a separate predictor of fatal stroke in individuals with essential hypertension	The study only included hypertension patients without cardiovascular illness at entrance, limiting its applicability to the broader community
[8]	To ascertain if morning or evening home blood pressure measurements have a stronger correlation with the occurrence of stroke events in the general population	D: Ohasama study S: Prospective cohort study M: The Cox regression model	Home blood pressure readings taken in the morning and evening are both highly correlated with the risk of stroke, although morning readings are more likely to indicate a stroke is imminent	The study's dependability may be compromised due to errors or variability in data collection due to the accuracy of participants following protocols for home blood pressure measures
[82]	To determine if atorvastatin reduces stroke risk in diabetics and if predictive indicators can identify stroke risk	D: CARDS trial S: Randomized controlled trial M: Cox regression models	Administering atorvastatin has a notable effect in decreasing the likelihood of stroke in people with diabetes when compared to a placebo	The study exclusively targeted persons with diabetes, which may restrict the generality of the results to non-diabetic populations and those with varying health profiles
[81]	To determine if CAC may be used as a predictor of stroke events	D: Heinz Nixdorf recall cohort S: Prospective cohort study M: Cox proportional hazards regressions	CAC was an independent predictor of stroke, along with age, systolic blood pressure, and smoking, in multivariable analysis	The study's findings may not be applicable to individuals under 45 or 75 years old, or those with pre-existing coronary heart disease or myocardial infarction
[86]	To examine the multivariate connection between infarct magnitude, 6 clinical factors, and 3 months outcomes in ischemic stroke patients	D: TirilazadMesylate in acute stroke (RANTTAS) S: Prospective cohort design (outcome) M: Multivariable logistic regression techniques	A study of ischemic stroke patients' examination and 1 week infarct magnitude predicted outcomes better than individual predictors	The study's findings may not apply universally to all ischemic stroke patients due to its limited sample representation and potential overlooking of alternative stroke outcome predictors
[13]	To determine if the severity of ischemic brain damage on MR DWI can offer supplementary prognostic data to clinical variables	D: The Beth Israel Deaconess medical center S: Prospective observational design M: Logistic regression	Combining NIHSS score, time from stroke onset to MR DWI, and volume of ischemic brain tissue on MR DWI enhanced stroke recovery prediction	The study is limited by its relatively small sample size and the probable lack of generalizability of the findings

Table 9 (continued)

Studies	Objectives	Dataset, study types and methodology	Key findings	Limitations
[76]	To analyze whether DWI characteristics identified early after stroke commencement might predict disability, mortality, and neurological recovery	D: Western general hospital, Lothian University S: Prospective observational study (outcome) M: logistic regression model	DWI with larger lesion volumes, important brain areas, and significant diffusion restriction increased disability, death, and neurological recovery at 3 months	The study's limitations include a small sample size of 96 patients, a focus on early DWI features, and potential inaccuracies in addressing post-stroke brain changes
[183]	To develop and test in-hospital mortality prediction models for ischemic stroke using Get With the Guidelines–Stroke Program data	D: The Get With the Guidelines–Stroke database. S: Retrospective cohort study M: Logistic regression	The study developed a prediction model using GWTG—stroke Program data to predict in-hospital mortality risk after ischemic stroke, achieving a C statistic of 0.72	The GWTG-stroke sample may not be representative due to voluntary participation, limiting inclusion to institutions with specific interest in stroke quality improvement
[131]	To investigate the independent role of stroke location in predicting cognitive and functional outcomes in patients with ischemic stroke	D: Bordeaux University Hospital S: Prospective cohort study M: Multivariate logistic regression models	Stroke site significantly impacts cognitive outcomes three months post-stroke, enhancing predictions accuracy by including age, initial NIHSS scores, and stroke volume in multivariate models	The study's results may not be universally applicable due to its limited sample size and voxel based lesion symptom mapping, which may not fully cover stroke outcomes
[223]	To estimate 1 year ischemic stroke recurrence risk using clinical risk variables and retinal features	D: Shenzhen traditional Chinese medicine hospital S: Prospective cohort study M: Multivariate logistic model	The study demonstrated a sensitivity and specificity of 69.1% and 68.4% for clinical risk factors and 62.0% and 64.4% for retinal characteristics, respectively, enhancing prediction accuracy to 72.5% and 70.7%	The study's small sample size and single center design may limit applicability, and the lack of detailed procedures could affect reproducibility and long term effects
[129]	To assess if transcranial Doppler ultrasound can detect asymptomatic cerebral emboli in patients with carotid artery stenosis, potentially improving therapeutic intervention targeting	D: The Doppler audio signal S: Prospective cohort study M: Kaplan-Meier analysis	Asymptomatic embolization is a significant predictor of future ischemia episodes, with an adjusted odds ratio of 8.10, even after considering other risk factors	The study's limitations include a small sample size, observational design, and single facility, suggesting a comprehensive review to confirm transcranial Doppler ultrasound's effectiveness in identifying high-risk patients
[2]	To investigate the correlation between transcranial Doppler ultrasound detection of embolic signals and increased risk of stroke or TAI in individuals with asymptomatic carotid stenosis	D: The Asymptomatic Stenosis Embolus Detection Study S: Multicenter prospective cohort study M: Kaplan-Meier analysis and Cox proportional hazards regression analysis	Transcranial Doppler ultrasound identified embolic signals in asymptomatic carotid stenosis, however they did not significantly enhance the risk of future ipsilateral stroke or transient ischemic attack or survival without these events	The study's small sample size and lack of significant associations between embolic signals and future stroke or TIA in asymptomatic carotid stenosis patients suggest transcranial Doppler ultrasound may not be a reliable prognostic indicator
[47]	To determine the incidence and prognostic factors of aspiration occurring within a 5 days period following an acute stroke	D: The Veterans affairs medical center in New Orleans S: Case series M: Chi-square analyses and logistic regression	Logistic regression analysis demonstrated that the combination of aberrant volitional cough and cough with swallow accurately predicted aspiration with a precision of 78%	The study's limitations include a small sample size of 55 patients and its focus on acute stroke patients within 5 days
[14]	To determine the case fatality rate within 30 days and identify factors affecting patients' first ever stroke, including stroke type, pre-stroke functional status, and age	D: The Oxfordshire community stroke project S: Prospective, community based cohort study M: Chi-square test	The study found that 19% of stroke patients die within 30 days, with higher rates for pre-stroke functional dependency and age	The findings highlight the need for prompt intervention and tailored treatment. Its focus on a single community, lack of comprehensive functional abilities assessment, 30 days follow-up period, and insufficient data on stroke management

Appendix C: ML-Based Stroke Prediction Studies

See Table 10.

Table 10 An overview of various studies and their methodologies, data sources, performance, and limitations of machine learning-based models on stroke prediction

Study	Objective	Data source, study type, methodology, and performance	Limitation
[182]	To evaluate various techniques, we will examine a suggested model for predicting strokes using the CHS dataset	Data source (D): CHS Dataset Study type (S): Prediction Methodology (M): ANN Performance (P): 97.7% accuracy	The study's limitations include its reliance on the CHS dataset, which may not cover all demographic and risk variables related to stroke, and its inability to consider comprehensibility
[39]	To forecast the result of stroke by employing knowledge discovery process (KDP) techniques	D: TurgutOzal Medicine Center, Inonu University, Malatya S: Prediction M: ANN P: Accuracy 81.82% and AUC 0.928	The study's limitations include a small sample size of 297 individuals, a dataset exclusively from one emergency medicine department, and potential bias in the selection process
[51]	To identify the most efficient pair of features for predicting strokes and assess the accuracy of ML algorithms in making stroke predictions using these features	D: Stroke prediction dataset from Kaggle S: Prediction M: ANN P: Precision 0.80, Recall 0.74, F-Score 0.77, Accuracy 0.77	The model's accuracy is impeded by the absence of distinctive features in the dataset, and its current size is inadequate for training DNN
[15]	To develop a Stroke Prediction algorithm using ML	D: Stroke Analysis Mendeley Data S: Prediction M: Improved RF P: Accuracy to 96.97%	Reliance on historical or simulated data for algorithm development and assessment, which may not fully encompass the complexity and diversity of clinical situations
[62]	To investigate stroke patient prognosis parameters and develops a ML prediction model to predict mortality and morbidity three months post-admission	D: The Stroke Unit of the Hospital Clinico Universitario of Santiago de Compostela (Spain) S: prospective observational study (Prediction) M: RF P: Over 0.90 AUC	The study's limitations include its reliance on a single tertiary hospital, potentially limiting its applicability to other healthcare settings and potentially introducing additional variables
[189]	To develop and evaluate ML models for predicting stroke incidence probabilities using physiological indicators	D: Stroke prediction dataset from Kaggle S: Prediction M: RF P: Accuracy 96%	Reliance on a single dataset, Stroke Prediction, potentially containing biases or limitations, emphasizing the need for larger datasets
[177]	To supplement current diagnosis techniques using a functional ANN model	D: 50 patients who have symptoms of stroke Disease S: Data prediction M: ANN P: accuracy of 89%	The study's limitations include a small sample size of 50 patients who have symptoms of stroke disease
[23]	To investigate if machine learning techniques can effectively predict SICH in patients with acute ischemic stroke after thrombolysis treatment, compared to established prognostic tools	D: Division of Brain sciences, Imperial College London, Charing cross hospital campus, Fulham Palace Rd., London S: Retrospective analysis M: SVM P: AUC 0.744	This study's retrospective nature and small sample size of 116 patients, including 16 SICH patients, may introduce bias or limit data collection and impact generalizability
[6]	To evaluate various medical data mining methods for predicting ischemic stroke	D: TurgutOzal Medical Centre, Inonu University, Malatya, Turkey S: Methodological investigation (Prediction) M: SVM P: Accuracy 0.9789, AUC 0.9783	A potential constraint of this study is the comparatively limited sample size, which may impact the capacity to apply the findings to a wider population
[94]	To evaluate the accuracy and life saving potential of three classification algorithms (DT, NB, and NN) in predicting stroke based on patient demographics	D: Faculty of Physical Therapy, Mahidol University, Thailand from 2012 to 2015 S: Prediction M: DT P: Accuracy 0.75, AUC 0.725, FP rate 0.15, FN rate 0.45	The study's emphasis on demographic data may constrain the precision of predictive models, as it fails to take into account thorough medical diagnoses or clinical data
[225]	To compare tree boosting and MLPs with traditional LR methods for stroke outcome prediction	D: The institutional ethics committee of CharitéUniversitätsmedizin Berlin S: Outcome prediction M: LR P: AUC 0.82	Concentrate on a particular dataset with a limited sample size

Table 10 (continued)

Study	Objective	Data source, study type, methodology, and performance	Limitation
[115]	To analyze stroke likelihood in patients with multiple health conditions using standard risk scores, a clinical index, and ML methods	D: A prospective US cohort of 3,435,224 patients from medical databases in a 2 years investigation S: Large scale prospective cohort study M: LR P: C Index 0.866, 95%CI 0.856–0.876	Despite limitations such as reliance on medical database data and observational design, is the largest prospective cohort assessing stroke risk in real world patients
[37]	To assess the efficacy of Cox models, ML algorithms, and ensemble models in predicting stroke risk in a prospective Chinese population	D: The China Kadoorie Biobank (CKB) S: prospective cohort study (Prediction) M: Ensemble approach P: Mean Accuracy: 78%, mean specificity: 78.5%, mean PPV: 25%	The study's stroke risk assessment is limited by AF and right censored data, and caution is advised when extrapolating findings to other populations
[169]	To develop a ML model that concentrate on physiological aspects to forecast the possibility of a brain stroke	D: 'Stroke prediction dataset' from Kaggle S: Predictive modelling study M: NB P: Accuracy 82%	Small data size and did not consider other potential risk factors like lifestyle choices or genetic predispositions that could contribute to stroke
[55]	To use ML methods to create a strong framework for long term stroke risk prediction, emphasizing early symptom identification and its correlation with stroke occurrence	D: 'Stroke prediction dataset' from Kaggle S: Prediction M: Stacking P: AUC 98.9%, accuracy 98%, precision 97.4%	Reliance on specific traits or variables in participant profiles, which may not fully encompass all potential stroke risk factors
[58]	To create a predictive model for stroke disease using ML techniques, considering 8 different factors	D: The medical clinic of Bangladesh S: Prediction M: Weight voting P: Accuracy 97%	The dependence on certain risk variables and features used in the prediction model, which might not encompass all potential predictors of stroke

Appendix D: Deep Learning and Hybrid Models for Stroke Prediction

See Table 11.

Table 11 An overview of various studies aimed at uncovering risk factors for stroke, their methodologies, key findings, and limitations of deep learning and hybrid models

Study	Objective	Data source, study type, methodology and performance	Limitation
[83]	To explore the potential of health insurance claims' extensive EMC data for predictive analytics on stroke occurrence, utilizing DNNs and other ML methods	Data source (D): The National Health Insurance Research Database (NHIRD) Study type (S): Prediction Methodology (M): DNN Performance (P): Accuracy 0.873, specificity 0.871, sensitivity 0.845, unweighted average recall 0.858	The complexity of DNN models, its reliance on EMC data, the need for significant computational resources, and the need for external validation
[116]	To improve cerebral stroke prediction using physiological data by developing a hybrid ML model to address data incompleteness and class imbalance issues	D: Stroke prediction dataset from Kaggle S: Prediction M: RF Regression and AutoHPO Based on DNN P: FNR 19.1%, FPR 33.1%, accuracy 71.6%, sensitivity 67.4% FNR 19.1%, FPR 33.1%, accuracy 71.6%, sensitivity 67.4%	The class imbalance and possible incompleteness of the dataset that might impact the generalizability of the model
[34]	To identify stroke using DL model that use predictive analytical techniques to identify stroke in heart illness dataset	D: UCI ML, Heart disease dataset S: Prediction M: DL (feed forward multi-layer ANN) P: Mean square error (MSE) is 0.2596, Mean value 36.73%, Standard deviation 0.084	Heart disease specific dataset, the need for extensive computational resources, and the lack of interpretability in DL models, which may not be universally applicable
[138]	To develop a predictive model using CNNdeep to accurately predict lesion volume in patients with acute IS, improving volume assessment accuracy and personalized treatment planning	D: I-KNOW multicenter (105) and remote ischemic preconditioning (117) studies S: comparative analytical study (predicting outcome) M: CNNdeep P: AUC = 0.88 ± 0.12	Small sample size and the intricacy of the CNNdeep model may limit its applicability in specific clinical contexts
[221]	To develop a stroke prediction system that utilizes real time biodata, specifically EMG signals, and AI techniques	D: Chungnam National University Hospital S: Prediction M: LSTM P: Accuracy 98.958%	The need for additional validation in larger, diversified datasets and its suitability in clinical situations
[218]	To explore the potential of DL techniques in predicting strokes using ECG data as a diagnostic tool	D: AUSA Shenzhen Inc S: Prediction M: DenseNet P: Accuracy 85.82%	The need for further validation with larger, more diverse datasets and considering confounding variables in ECG interpretation for stroke prediction
[36]	To develop a non-intrusive, instantaneous stroke forecasting algorithm using unprocessed EEG data, overcoming the cost and convenience of conventional imaging methods	D: Chungnam National University Hospital S: Prediction M: CNN-Bidirectional LSTM P: Accuracy 94%, FPR 6.0%, FNR 5.7%	The need for larger dataset validation, practical implementation, and long term effectiveness comparison in clinical settings, requiring advanced processing capabilities
[222]	To develop a real time stroke prediction system using ECG and PPG data to identify early signs of stroke, enabling timely medical intervention	D: Chungnam National University Hospital, Republic of Korea S: Prediction M: CNN-LSTM P: Accuracy 99.15%	The need for larger dataset validation and real time data collection, which may not be accessible in all healthcare settings
[35]	To use EHRs to predict stroke using RNN and LSTM hidden units	D: Department of Medical Services, The Ministry of Public Health of Thailand S: Prediction M: LSTM-RNN P: Accuracy range 0.9279–0.9998, F1-score range 0.9626–0.9999	The need for large datasets for accurate prediction results, and further validation with larger, varied datasets is needed for model robustness
[12]	To develop DL models that utilize clinical data and CT brain images to predict individual risk benefit ratios for intravenous thrombolysis decision making	D: Tertiary hospitals in South Australia S: Prediction M: CNN + ANN P: mRS90 Accuracy 0.74, F1 score 0.69 and for NIHSS24 accuracy 0.71, F1 Score 0.74	Small sample size and reliance on clinical data and noncontrast CT scans may limit its applicability, and further research is needed to validate the models in diverse clinical environments

Author Contributions M.S.S. conceived the research idea, designed the study framework, and led the manuscript writing. He conducted the systematic literature review, structured the comparative analysis, and coordinated all revisions. K.T. assisted with the review methodology and contributed to drafting the related work and background sections. P.K.B. was responsible for compiling the dataset references, preparing Figs. 1–5 and Tables 1–4, and analyzing performance metrics. P.C. provided overall supervision, reviewed the manuscript critically for domain relevance, and contributed to refining the discussion and conclusion. All authors reviewed and approved the final manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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